

University of Niagara Falls
Data Analytics MSc



Data Analytics Case Study 1 (DAMO-501-22)

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Chapter 1. Initial Data Understanding

1.1 Dataset Context

The Calgary Building Inspections dataset contains administrative records documenting the permitting and inspection processes for construction activities within the City of Calgary. Each permit represents a construction-related activity, such as building, renovation, or structural modification, that must comply with municipal regulations. Throughout the permit lifecycle, inspectors conduct site visits, record inspection outcomes, and document deficiencies that must be resolved before the permit can be closed.

Because the dataset originates from real municipal operations, it includes typical administrative challenges such as missing fields, inconsistent entries, duplicated descriptions, and varying levels of detail. These characteristics make the dataset valuable for analytical work, hypothesis development, and SQL-based data extraction, while also requiring careful attention to data quality and limitations.

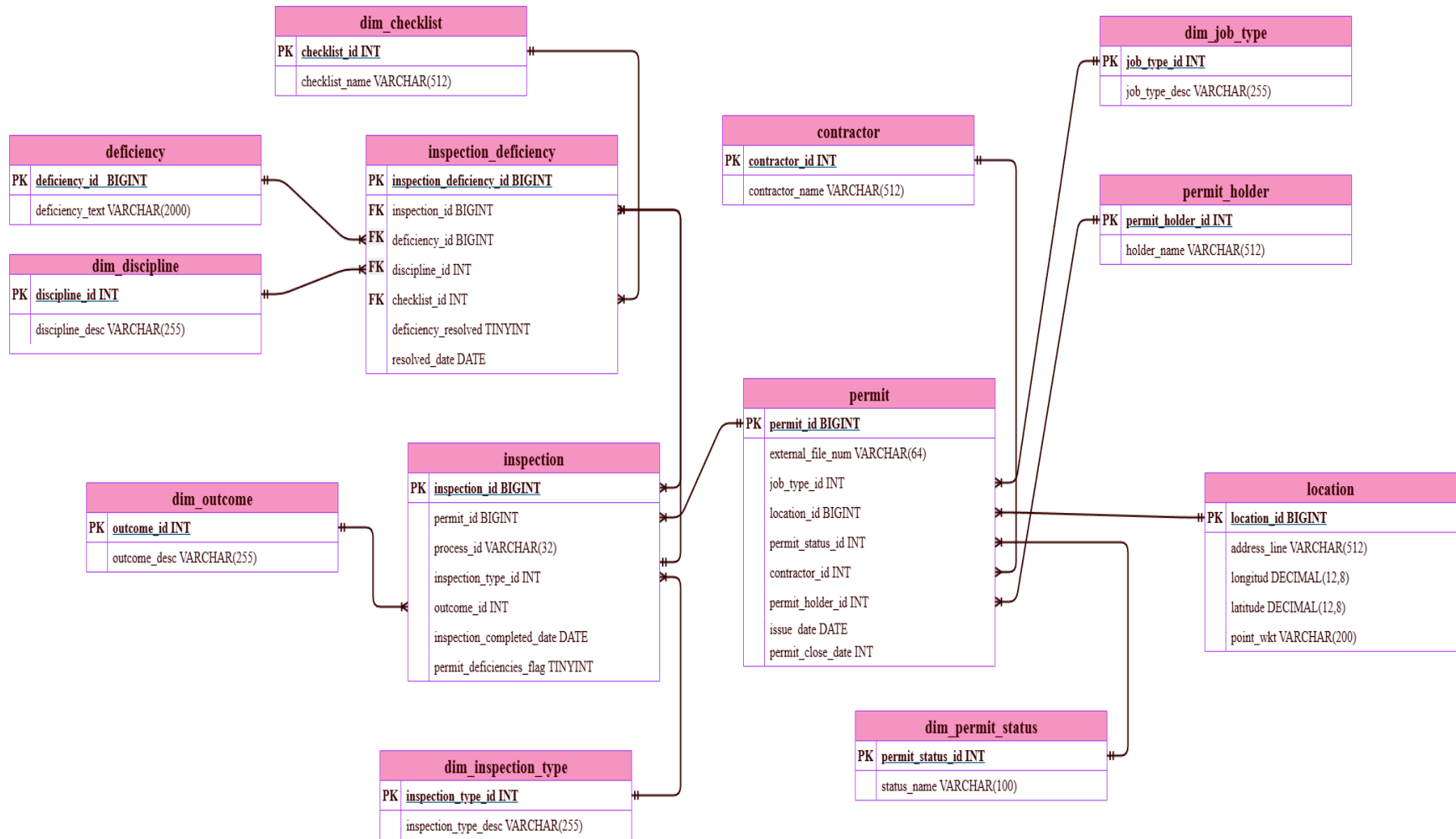
The relational structure of the database enables multi-table joins and supports a wide range of analytical tasks, including examining inspection outcomes, identifying deficiency patterns, evaluating contractor performance, comparing job types, and analyzing permit durations. These capabilities directly support the development of a meaningful research question.

1.2 Database Schema Overview

A clear definition of the entities and their corresponding attributes is provided in the following list, and an entity-relationship diagram (ERD) created using **draw.io** is included in the report to visually illustrate how the tables relate to one another. The dataset consists of multiple interconnected tables that form a relational structure suitable for analytical work:

1. **Permit:** Contains permit-level information such as `issue_date`, `permit_close_date`, `job_type_id`, `permit_status_id`, `contractor_id`, `permit_holder_id`, and `location_id`. This table is the central reference point for most analyses.
2. **Inspection:** Records inspection events linked to each permit, including `inspection_type_id`, `outcome_id`, `inspection_completed_date`, and a deficiency flag.

3. **Inspection_Deficiency:** A junction table linking inspections to deficiencies, containing discipline_id, checklist_id, deficiency_resolved, and resolved_date.
4. **Deficiency:** Stores standardized textual descriptions of deficiencies.
5. **Location:** Provides address information and geographic coordinates for each permit site.
6. **Contractor:** Contains contractor names for performance analysis.
7. **Permit_Holder:** Identifies the individuals or organizations responsible for each permit.
8. **Dimension Tables:** (dim_job_type, dim_permit_status, dim_inspection_type, dim_outcome, dim_discipline, dim_checklist) provide descriptive labels for categorical identifiers.
9. This relational structure supports detailed analysis of relationships among permits, inspections, outcomes, deficiencies, contractors, and locations.



Entity-Relationship Diagram

Calgary

1.3 Initial Exploratory Analysis Through Business-Driven Questions

To develop a meaningful understanding of the Calgary Building Inspections dataset, an initial exploratory analysis was conducted using a set of business-oriented questions. These questions reflect operational concerns relevant to city administrators, contractors, homeowners, and the public. *For each question, the corresponding SQL code is provided along with the resulting output table. When a query produces many records, only the first ten rows are displayed to ensure clarity and readability.* Each question is followed by a concise interpretation of the results, highlighting patterns that inform the analytical direction of the project.

1.3.1 Question 1: Which types of construction projects are most common in Calgary?

Understanding the distribution of job types provides insight into where the city allocates most of its permitting and inspection resources.

```
SELECT j.job_type_desc AS Job_Type_Description,  
       COUNT(*) AS Frequency  
FROM permit p  
JOIN dim_job_type j ON p.job_type_id = j.job_type_id  
GROUP BY j.job_type_desc  
ORDER BY Frequency DESC;
```

Job_Type_Description	Frequency
Single Construction Permit	39305
Residential Improvement Project	25354
Commercial / Multi Family Project	13732
(unknown)	3

Key Insights

1. **Single Construction Permits dominate** the dataset with **39,305 records**, indicating that most municipal activity is tied to individual residential builds.
2. **Residential Improvement Projects** are the second-largest category (**25,354 permits**), reinforcing that residential work drives most of the inspection demand.

3. **Commercial / Multi-Family Projects** appear far less frequently (**13,732 permits**), but these projects are typically more complex and may require more inspections per permit.
4. The **(unknown)** category contains only **3 records**, suggesting minimal but present data entry or classification issues.

1.3.2 Question 2: Do some project types require more inspections than others?

Inspection frequency serves as an indicator of project complexity and potential delays.

```
SELECT j.job_type_desc AS Job_Type_Description,
       AVG(i_count) AS Average_Inspections
FROM (
  SELECT permit_id, COUNT(*) AS i_count
  FROM inspection
  GROUP BY permit_id
) AS x
JOIN permit p ON x.permit_id = p.permit_id
JOIN dim_job_type j ON p.job_type_id = j.job_type_id
GROUP BY j.job_type_desc;
```

Job_Type_Description	Average_Inspections
Single Construction Permit	1.5279
Residential Improvement Project	1.1057
Commercial / Multi Family Project	1.3814
(unknown)	2.6667

Key Insights

1. **Single Construction Permits require the highest average number of inspections (1.53)** among the major categories, suggesting these projects involve more checkpoints than typical residential improvements.
2. **Commercial / Multi-Family Projects also show a relatively high inspection average (1.38)**, consistent with their greater structural and regulatory complexity.

3. **Residential Improvement Projects require the fewest inspections on average (1.11)**, indicating these projects are simpler and more standardized.
4. The **(unknown)** category shows an unusually high average (2.67), which is likely due to **very low sample size (n=3)** rather than true project complexity — a small data quality flag.
5. Overall, the differences in inspection averages suggest that **project type influences inspection workload**, which may later relate to differences in permit duration.

1.3.3 Question 3: What are the most common inspection outcomes?

Inspection outcomes influence project timelines and highlight areas where compliance issues may arise.

```
SELECT d.outcome_desc AS Inspection_Outcome,
       COUNT(*) AS Frequency
FROM inspection i
JOIN dim_outcome d ON i.outcome_id = d.outcome_id
GROUP BY d.outcome_desc
ORDER BY Frequency DESC;
```

Inspection_Outcome	Frequency
Not Acceptable	38792
Acceptable with Conditions	33967
Acceptable	12255
Acceptable Add Inspection	11195
Acceptable - Close Permit	9046
Not Required	1301
Acceptable with Minor Deficiencies	127
Office Procedure	124
No Inspection	69
Unable to enter	59

Key Insights

1. **“Not Acceptable” is the most frequent outcome (38,792 inspections)**, indicating that a large portion of inspections initially fail to meet required standards. This suggests substantial rework and follow-up activity across the city.
2. **“Acceptable with Conditions” is also very common (33,967 inspections)**, showing that many inspections pass only partially and still require corrective actions before full approval.
3. Fully successful inspections, **“Acceptable” (12,255)** and **“Acceptable – Close Permit” (9,046)**, occur far less often, reinforcing that most inspections involve some level of deficiency or follow-up.
4. **“Acceptable Add Inspection” (11,195)** highlights that many projects require additional verification steps, further increasing inspection workload.
5. Rare outcomes such as **“Unable to Enter,” “No Inspection,”** and **“Office Procedure”** appear infrequently, but they indicate operational interruptions or administrative handling rather than field inspection results.
6. Overall, the distribution shows that **many inspections do not pass cleanly on the first attempt**, which has implications for project timelines, resource allocation, and ultimately permit duration.

1.3.4 Question 4: Are deficiencies more common in certain types of projects?

Deficiencies require rework and additional inspections, contributing to longer processing times.

```
SELECT j.job_type_desc AS Job_Type_Description,
       COUNT(*) AS Deficiency_Count
FROM inspection_deficiency id
JOIN inspection i ON id.inspection_id = i.inspection_id
JOIN permit p ON i.permit_id = p.permit_id
JOIN dim_job_type j ON p.job_type_id = j.job_type_id
GROUP BY j.job_type_desc
ORDER BY Deficiency_Count DESC;
```

Job_Type_Description	Deficiency_Count
Single Construction Permit	69806
Residential Improvement Project	32262
Commercial / Multi Family Project	21725
(unknown)	9

Key Insights

1. **Single Construction Permits show the highest number of deficiencies (69,806)**, which aligns with their high permit volume. This indicates that residential new-builds generate the largest share of rework and follow-up activity.
2. **Residential Improvement Projects also show a substantial number of deficiencies (32,262)**, suggesting that renovation-type work frequently triggers corrective actions.
3. **Commercial / Multi-Family Projects have 21,725 deficiencies**, which is lower in absolute terms but proportionally high relative to their smaller permit volume. This implies **higher complexity and more issues per project**.
4. The **(unknown)** category has only **9 deficiencies**, which is expected given its extremely small sample size, but it still signals minor classification inconsistencies.
5. Overall, the distribution shows that **deficiencies are widespread across all project types**, reinforcing that inspection rework is a major operational factor that may influence permit duration.

1.3.5 Question 5: What is the distribution of inspections per permit?

Examining the distribution of inspection counts helps identify variability and potential outliers.

```
SELECT i_count AS Inspections_Per_Permit,
```

```
    COUNT(*) AS Frequency
```

```
FROM (
```

```
    SELECT permit_id, COUNT(*) AS i_count
```

```
    FROM inspection
```

```
    GROUP BY permit_id
```

```
) AS x
```

GROUP BY i_count
ORDER BY i_count
LIMIT 10;

Inspections_Per_Permit	Frequency
1	57448
2	15270
3	4243
4	1052
5	260
6	68
7	28
8	10
9	5
10	5

Key Insights

1. Most permits require **only one inspection (57,448 permits)**, showing that most projects are simple and follow a straightforward approval process.
2. A significant number of permits require **two inspections (15,270)**, indicating that follow-up checks are common but still manageable.
3. Higher inspection counts drop off sharply: **3 inspections (4,243)**, **4 inspections (1,052)**, and **5+ inspections** become increasingly rare.
4. The long tail, permits requiring **6 to 10 inspections**, represents a small subset of projects that likely involve **complex issues, repeated deficiencies, or multi-stage verification**.
5. Overall, the distribution is heavily right-skewed, confirming that **most permits are low-complexity**, while a small number demand **disproportionate inspection effort**.

1.3.6 Question 6: What are the summary statistics for permit duration?

Summary statistics provide an early indication of how long permits typically remain open.

SELECT

ROUND(MIN(DATEDIFF(permit_close_date, issue_date)), 2) AS Minimum,
ROUND(AVG(DATEDIFF(permit_close_date, issue_date)), 2) AS Average,
ROUND(MAX(DATEDIFF(permit_close_date, issue_date)), 2) AS Maximum,

```

ROUND(STDDEV(DATEDIFF(permit_close_date, issue_date)), 2) AS Std_Deviation
FROM permit
WHERE permit_close_date IS NOT NULL;

```

Minimum	Average	Maximum	Std_Deviation
0	356.41	2916637	14948.97

Key Insights

1. The **minimum duration is 0 days**, meaning some permits were closed immediately or on the same day.
2. The **average duration is 356.41 days**, showing that most permits stay open for roughly one year.
3. The **maximum value (2,916,637 days)** is clearly an extreme outlier caused by incorrect or missing dates.
4. The **very large standard deviation (14,948.97)** confirms that the dataset contains major inconsistencies or abnormal records.

1.3.7 Question 7: Do inspection outcomes differ by job type?

Cross-tabulating outcomes by job type reveals whether certain project categories experience more compliance challenges. [Table A1](#) presents the complete dataset used for the analysis.

```

SELECT j.job_type_desc AS Job_Type_Description,
       d.outcome_desc AS Inspection_Outcome,
       COUNT(*) AS Frequency
FROM inspection i
JOIN permit p ON i.permit_id = p.permit_id
JOIN dim_job_type j ON p.job_type_id = j.job_type_id
JOIN dim_outcome d ON i.outcome_id = d.outcome_id
GROUP BY j.job_type_desc, d.outcome_desc
ORDER BY j.job_type_desc, Frequency DESC;

```

Job_Type_Description	Inspection_Outcome	Frequency
(unknown)	Acceptable Add Inspection	6
(unknown)	Not Acceptable	2

Commercial / Multi Family Project	Acceptable Add Inspection	8826
Commercial / Multi Family Project	Not Acceptable	3872
Commercial / Multi Family Project	Acceptable	3718
Commercial / Multi Family Project	Acceptable with Conditions	2436
Commercial / Multi Family Project	Acceptable with Minor Deficiencies	35
Commercial / Multi Family Project	No Inspection	20
Commercial / Multi Family Project	Not Required	16
Commercial / Multi Family Project	Incorrectly called for	13

Key Insights

1. Single Construction Permits drive most inspection failures
 - a. **Not Acceptable** outcomes dominate this category (**27,970 cases**), showing that new residential builds generate the highest volume of failed inspections.
 - b. They also have very high **Acceptable with Conditions** counts (**23,947**), meaning many inspections pass only partially.
2. Residential Improvement Projects show a more balanced pattern
 - a. They have a strong number of **successful closures** (**9,046 “Acceptable – Close Permit”**), more than any other category.
 - b. Still, **Not Acceptable** outcomes remain high (**6,948**), indicating frequent deficiencies in renovation work.
3. Commercial / Multi-Family Projects require more follow-up
 - a. **Acceptable Add Inspection** is extremely common (**8,826**), suggesting multi-stage verification and higher complexity.
 - b. They also show a notable number of **Not Acceptable** outcomes (**3,872**), reinforcing the complexity of commercial builds.
4. Rare outcomes highlight operational issues
 - a. Low-frequency outcomes like **Unable to Enter**, **Incorrectly Called For**, **Stop Work**, and **Office Procedure** appear across categories, indicating occasional scheduling, access, or administrative problems.
5. The (unknown) category is tiny but inconsistent

- a. Only **8 total inspections**, but mixed outcomes — a small data-quality flag.

1.3.8 Question 8: Are missing close dates more common in certain project types?

Missing close dates affect the ability to calculate permit duration and may indicate incomplete or ongoing projects.

```
SELECT j.job_type_desc AS Job_Type_Description,
       COUNT(*) AS Missing_Close_Dates
FROM permit p
JOIN dim_job_type j ON p.job_type_id = j.job_type_id
WHERE p.permit_close_date IS NULL
GROUP BY j.job_type_desc;
```

Job_Type_Description	Missing_Close_Dates
Single Construction Permit	526
Commercial / Multi Family Project	169
Residential Improvement Project	1502
(unknown)	1

Key Insights

1. Residential Improvement Projects have the highest number of missing close dates (**1,502**), suggesting inconsistent data entry or delays in administrative closure for renovation-type permits.
2. Single Construction Permits also show a notable number of missing close dates (**526**), which may affect duration calculations and skew averages if not handled.
3. Commercial / Multi-Family Projects have fewer missing close dates (**169**), indicating better closure tracking or more structured project workflows.
4. The **(unknown)** category has only **1** missing date, reinforcing that this category is extremely small but still contains minor data quality issues.
5. Overall, the distribution shows that missing close dates are a significant data quality concern, especially for residential projects, and must be addressed before performing accurate duration analysis.

1.3.9 Conclusion

Overall, the exploratory queries revealed clear patterns in inspection activity, deficiency prevalence, and permit behavior across job types. Single Construction Permits and Residential Improvement Projects generated the highest volumes of inspections and deficiencies, while Commercial / Multi-Family Projects showed greater complexity through higher rates of follow-up inspections. Inspection outcomes were dominated by “Not Acceptable” and “Acceptable with Conditions,” indicating that most inspections require some level of rework. The analysis also highlighted several data quality issues, including missing close dates, inconsistent outcome categories, unresolved deficiencies, and extreme outliers in permit duration. These limitations are typical of administrative datasets and must be considered when interpreting trends, as they may influence workload estimates, performance metrics, and subsequent analytical conclusions.

1.4 Identification of Data Quality Issues, Assumptions, and Limitations

The initial exploration revealed several data quality issues that must be acknowledged before conducting deeper analysis. Missing or incomplete fields were common, particularly in *permit_close_date*, *inspection_completed_date*, and *resolved_date*, which directly affect duration calculations and the ability to track deficiency resolution. Some permits contained extreme or unrealistic values, such as multi-million-day durations, indicating incorrect or defaulted dates. Deficiency descriptions also showed inconsistencies, including duplicated labels, minor spelling variations, and unresolved deficiencies lacking timestamps. Additionally, inspection outcomes included low-frequency categories such as “Unable to Enter,” “Incorrectly Called For,” and “CSR Skipped,” suggesting operational or administrative irregularities. These limitations are typical of real-world administrative datasets and introduce uncertainty into the analysis. As a result, all findings must be interpreted with caution, recognizing that data gaps and inconsistencies may influence observed patterns and statistical results.

1.5 Preliminary Observations That Motivate the Analytical Direction

Early findings highlight several patterns that guide the direction of the analysis. Single Construction Permits and Residential Improvement Projects generate the highest volumes of inspections and deficiencies, indicating that these categories drive most operational workload.

Commercial / Multi-Family Projects, while fewer in number, show higher rates of follow-up inspections, suggesting greater complexity and multi-stage verification. Inspection outcomes are dominated by “Not Acceptable” and “Acceptable with Conditions,” revealing that most inspections require corrective action rather than passing cleanly. The distribution of inspections per permit is heavily right-skewed, with most permits requiring only one inspection but a small subset demanding extensive rework. Missing close dates are especially prevalent in Residential Improvement Projects, which may distort duration metrics if not addressed. Together, these observations point toward a deeper investigation of how job type, inspection behavior, and deficiency patterns influence permit duration and operational efficiency.

Chapter 2. Problem Definition and Research Question

2.1 Problem Definition

The City of Calgary's permitting system records detailed information on construction projects, including permit issue dates, inspection outcomes, and deficiency activity. A key performance indicator for both industry stakeholders and municipal staff is **permit duration**, defined as the number of days between permit issuance and closure. Preliminary exploration of the dataset shows substantial variation in these durations across job types, suggesting that different categories of projects may experience different regulatory timelines.

Commercial projects often involve more complex building requirements, such as fire protection systems, mechanical reviews, and accessibility standards, while residential projects tend to follow more standardized processes. These differences may influence the number of inspections required, the frequency of deficiencies, and ultimately the time needed to close a permit. This study focuses on determining whether these operational differences translate into **measurable differences in permit duration**.

2.2 Research Question

Do commercial building permits take longer to close than residential building permits?

This question is directly measurable using the dataset, which includes permit issue dates, close dates, and job type classifications. It allows for a clear statistical comparison between two well-defined groups.

2.3 Justification of Relevance

Understanding whether permit duration differs by job type is important for operational planning, resource allocation, and stakeholder expectations. Longer processing times can delay construction schedules, increase project costs, and affect customer satisfaction. For the City of Calgary, identifying which project types consistently require more time can support improvements in inspection workflows, staffing decisions, and service delivery standards. Because the dataset contains all necessary fields to calculate permit duration and categorize job types, the research question is both relevant and well-supported by the available data.

Chapter 3. Hypotheses Formulation

3.1 Hypothesis Writing

The purpose of this hypothesis test is to determine whether the average permit duration differs between residential and commercial construction projects in the City of Calgary. Permit duration is measured as the number of days between the permit issue date and the permit close date.

Let:

$u(\text{commercial})$ = mean permit duration for commercial permits

$u(\text{residential})$ = mean permit duration for residential permits

Null Hypothesis (H_0)

H_0 : The null hypothesis states that there is no significant difference in the mean permit duration between residential and commercial permits, i.e., $u(\text{commercial}) = u(\text{residential})$

Alternative Hypothesis (H_1)

H_1 : The alternative hypothesis states that commercial permits take longer than residential permits on average, i.e., $u(\text{commercial}) > u(\text{residential})$

3.2 Hypothesis Explanation & Connection to Data

Commercial projects typically involve more complex regulatory requirements—such as specialized mechanical systems, fire protection reviews, and multi-stage inspections—while residential projects follow more standardized processes. These operational differences suggest that commercial permits may require more time to close. The hypothesis therefore reflects a realistic expectation grounded in construction workflows and inspection practices.

To ensure the hypothesis is **testable and statistically valid**, the following data-handling decisions will be applied:

1. Only closed permits will be included.

Permit duration requires both an issue date and a close date. Records missing a close date will be excluded because they cannot produce a valid duration.

2. Outliers will be identified using the IQR method with a 5% threshold.

Outliers will be detected separately for residential and commercial permits using the classical **1.5 × IQR rule**. For each group, we will compute:

- $Q1$
- $Q3$
- $IQR = Q3 - Q1$
- Lower bound = $Q1 - 1.5 \times IQR$
- Upper bound = $Q3 + 1.5 \times IQR$

If **more than 5%** of observations fall outside these bounds for either group, we will not simply remove them. Instead, we will apply the next step.

3. If the 5% threshold is exceeded, we will select a recent three-year period where the threshold is satisfied.

Because the full dataset spans **2000 to 2025**, some permits show extremely long durations (e.g., permits issued in early 2000 but closed in 2025). These durations are not “millions of days,” but they can reach **several thousand days**, which is far beyond typical permit timelines.

To avoid distortion:

- We will identify a **three-year window** where outliers represent **less than 5%** of observations for both residential and commercial permits.
- We will select the **most recent** such period to ensure relevance to current permitting practices.

This approach preserves data integrity while preventing extreme historical cases from biasing the statistical test.

4. Duration will be calculated consistently.

Duration = close_date – issue_date in days, using cleaned and validated date fields. These steps ensure that the comparison between residential and commercial permits is based on **complete**,

reliable, and methodologically consistent data, strengthening the validity of the hypothesis test.

Chapter 4. Data Collection and SQL Queries

This chapter presents the SQL procedures used to extract and prepare the dataset required to compare permit durations between residential and commercial projects. Short SQL queries (under ~10 lines) are shown directly in this chapter along with a preview of their output (first 10 rows). Longer queries are included in **Appendix B SQL Scripts**.

4.1 Initial Data Extraction Using SQL Techniques

4.1.1 Row Counts Across All Tables

Purpose: To obtain an overview of the size of each table in the *calgary_inspections* database and confirm the expected relational structure.

SQL Code: Because the row-count query exceeds 10 lines, it is included in **Appendix B ([Script B1](#))**.

Table	Row Counts
contractor	30
deficiency	81982
dim_checklist	131
dim_discipline	3
dim_inspection_type	14
dim_job_type	4
dim_outcome	20
dim_permit_status	13
inspection	107067
inspection_deficiency	123802
location	71771
permit	78394
permit_holder	348

Key Insights

The largest tables are *inspection*, *inspection_deficiency*, and *permit*, confirming that these are the core transactional tables used for analytical processing. Dimension tables contain far fewer records, consistent with a star-schema design.

4.1.2 DISTINCT Queries

Purpose:

To identify categorical variables and verify the number of unique values within key dimension tables. This step helps confirm category consistency and ensures that fields such as job type, inspection type, and outcome are suitable for grouping in later analysis.

SQL Code:

The DISTINCT queries applied across all tables exceed 10 lines and are therefore included in Appendix B ([Script B2](#)). As an example, we apply this to the dim_job_type table

job_type_id	job_type_desc
4	(unknown)
3	Commercial / Multi Family Project
2	Residential Improvement Project
1	Single Construction Permit

Key Insights

The DISTINCT extraction confirms that job type categories are limited and clearly defined. This low cardinality supports grouping permits into **Residential** and **Commercial**, which directly aligns with the research question. Similar checks across other dimension tables verified that categorical fields are consistent and appropriate for statistical analysis.

4.1.3 Table Structure Documentation

Purpose: To document the schema design, identify primary and foreign keys, and understand how tables relate to each other. Examining table structures ensures that joins between permits, inspections, deficiencies, and dimension tables are performed correctly during dataset construction.

SQL Code: Because the DESCRIBE statements were executed across all tables and exceed 10 lines, they are included in **Appendix B** ([Script B3](#)).

Field	Type	Null	Key	Default	Extra
permit_id	bigint	NO	PRI	NULL	auto_increment
external_file_num	varchar(64)	NO	UNI	NULL	
job_type_id	int	NO	MUL	NULL	
location_id	bigint	NO	MUL	NULL	
permit_status_id	int	NO	MUL	NULL	
contractor_id	int	YES	MUL	NULL	
permit_holder_id	int	YES	MUL	NULL	
issue_date	date	YES		NULL	

permit_close_date	date	YES		NULL	
-------------------	------	-----	--	------	--

Key Insights

The DESCRIBE outputs confirm that the database follows a **relational star-schema structure**, with transactional tables (permit, inspection, inspection_deficiency) linked to dimension tables through identifier fields such as job_type_id, inspection_type_id, and permit_status_id. This structure supports accurate joins and ensures that analytical variables—such as permit duration and job type—can be reliably extracted.

4.1.4 Missing Value Identification

Purpose: To evaluate data completeness before constructing the analytical dataset. Identifying missing values ensures that variables such as permit duration, job type, and inspection outcomes are based on complete and reliable records.

SQL Code: The missing-value queries were executed across all tables and exceed 10 lines; therefore, they are included in **Appendix B (Script B4)**. As an example permit table.

Table	permit_status_id	contractor_id	permit_holder_id	issue_date	permit_close_date
permit	0	78362	70422	43	2198

Key Insights

Missing values were identified across several tables, with the most notable being **missing permit_close_date values** in the *permit* table. These missing close dates were later excluded during data cleaning because permit duration cannot be calculated without them. The results of this step directly informed the filtering rules applied in the analytical dataset.

4.2 Utilization of appropriate SQL commands to extract relevant data from different tables

4.2.1 Duration Calculation

Purpose: To calculate the processing duration of each permit by measuring the number of days between the permit issue date and the permit close date. This variable is essential for comparing

processing times between residential and commercial permits and forms the core metric used in the statistical analysis.

SELECT

issue_date,

permit_close_date,

DATEDIFF(permit_close_date, issue_date) **AS** Duration_Days

FROM permit

LIMIT 10;

issue_date	permit_close_date	Duration_Days
2005-02-03	2005-05-31	117
2005-01-26	2006-06-20	510
2005-02-09	2008-05-14	1190
2005-06-30	2009-07-24	1485
2005-02-11	2006-10-11	607
2005-06-02	2005-10-14	134
2005-02-16	2005-06-07	111
2006-06-29	2007-06-18	354
2006-04-19	2006-10-20	184
2006-06-03	2007-01-31	242

Key Insights

The resulting variable **Duration_Days** represents the total number of calendar days required to complete each permit. This transformation converts raw date fields into a measurable quantitative metric, enabling descriptive statistics, outlier detection, and hypothesis testing.

Duration_Days becomes the primary variable used to compare processing times between residential and commercial permits.

4.2.2 Job Type Join

Purpose: To enrich the permit dataset by integrating job type classifications. This join allows detailed job type descriptions to be grouped into broader analytical categories, **Residential** and **Commercial**, which are required for comparing permit processing durations across project types.

SELECT

A.issue_date,

```

A.permit_close_date,
DATEDIFF(A.permit_close_date, A.issue_date) AS Duration_Days,
CASE
    WHEN B.job_type_desc = 'Residential Improvement Project' THEN 'Residential'
    WHEN B.job_type_desc = 'Commercial / Multi Family Project' THEN 'Commercial'
END AS Job_Type_Description
FROM permit A
JOIN dim_job_type B
    ON A.job_type_id = B.job_type_id
LIMIT 10;

```

issue_date	permit_close_date	Duration_Days	Job_Type_Description
2013-07-09	2015-09-11	794	NULL
2015-05-07	2018-10-15	1257	NULL
2022-11-17	NULL	NULL	NULL
2000-06-01	2000-10-13	134	Commercial
2001-10-19	2001-11-20	32	Commercial
2000-03-13	2000-09-07	178	Commercial
2000-11-21	2001-02-01	72	Commercial
2000-08-14	2000-10-05	52	Commercial
2001-04-30	2001-06-06	37	Commercial
2000-08-25	2002-02-06	530	Commercial

Key Insights

This transformation produces an enriched dataset that combines operational permit information with descriptive job classifications. The new variable **Job_Type_Description** simplifies the original job type categories into two groups—Residential and Commercial—making them directly comparable. At the same time, **Duration_Days** provides the primary quantitative measure used to evaluate differences in permit processing times. Together, these variables create a structured dataset suitable for descriptive analysis and hypothesis testing.

4.2.3 Filtering Valid Permits

Purpose: To retain only valid permit records that meet the analytical criteria required for comparing processing durations. This filtering step ensures that the dataset includes only

complete, relevant, and closed permits within the selected study period (2022–2024), and that job types are restricted to the two categories used in the research question: **Residential** and **Commercial**.

SELECT

CASE

WHEN B.job_type_desc = 'Residential Improvement Project' **THEN** 'Residential'

WHEN B.job_type_desc = 'Commercial / Multi Family Project' **THEN** 'Commercial'

END AS Job_Type_Description,

COUNT(*) **AS** Number_of_Permits_by_Job_Type

FROM permit A

INNER JOIN dim_job_type B

ON A.job_type_id = B.job_type_id

WHERE YEAR(issue_date) **BETWEEN 2022 AND 2024**

AND issue_date **IS NOT NULL**

AND permit_close_date **IS NOT NULL**

AND A.permit_status_id = **1**

AND B.job_type_id **IN** (2,3)

GROUP BY B.job_type_desc;

Job_Type_Description	Number_of_Permits_by_Job_Type
Residential	2826
Commercial	324

Key Insights

This filtering step produces a refined dataset containing only complete, valid, and analytically relevant permit records. By restricting the data to permits issued between 2022 and 2024, ensuring both issue and close dates are present, and selecting only completed permits within the Residential and Commercial job type categories, the resulting dataset becomes consistent with the research objective. This process removes incomplete or irrelevant observations, preserves

only records with measurable durations, and creates a clean foundation for comparing processing times between residential and commercial permits in subsequent statistical analysis.

4.2.4 Final Dataset Creation

Purpose: The final analytical dataset was constructed by integrating permit information, job type classifications, permit status, and calculated permit duration. This step combines all prior extraction, transformation, and filtering procedures into a single query (included in Appendix B SQL Script, [Script B5](#)) to produce a complete, analysis-ready dataset.

job_type_descripti on	status_nam e	issue_date	permit_close_da te	permit_duration_da ys
Residential	COMP	2022-01-10	2022-02-10	31
Residential	COMP	2022-03-30	2022-10-28	212
Commercial	COMP	2022-09-14	2023-02-22	161
Commercial	COMP	2022-12-07	2023-02-02	57
Commercial	COMP	2023-02-23	2025-03-28	764
Commercial	COMP	2022-11-01	2023-02-24	115
Residential	COMP	2022-04-29	2022-08-12	105
Residential	COMP	2024-04-22	2025-04-02	345
Residential	COMP	2022-05-03	2022-05-20	17
Residential	COMP	2022-05-03	2023-03-28	329

Key Insights

The resulting dataset includes the permit identifier, job type classification, permit status, issue date, close date, and the calculated duration in days. This final query transforms raw database records into an analysis-ready dataset by removing incomplete observations, restricting the study period to 2022–2024, selecting only valid and completed permits, classifying records into Residential and Commercial groups, and generating the quantitative variable *permit_duration_days*. Together, these steps ensure that the final dataset is clean, consistent, and aligned with the research objective of comparing processing times across job types.

4.2.5 Inspection Counts

Purpose: The purpose of this query is to quantify the total number of inspections recorded in the database. Establishing the overall inspection volume provides important context for understanding operational workload and supports later analyses involving inspection frequency, outcomes, and deficiency patterns.

```
SELECT COUNT(*) AS total_inspections
FROM inspection;
```

total_inspections
107067

Key Insights

This query returns a total of **107,067 inspections**, providing a clear baseline measure of inspection activity across all permits in the dataset. Establishing this count is important because it contextualizes subsequent analyses, such as yearly inspection trends, inspection outcomes, and deficiency rates, and ensures that later interpretations are grounded in an accurate understanding of the overall scale of inspection operations.

4.2.6 Total Inspections Grouped by Year

Purpose: The purpose of this query is to summarize inspection activity over time by calculating the total number of inspections completed each year. Grouping inspections by year provides insight into workload trends and supports later analyses involving inspection frequency, operational patterns, and potential seasonal or annual variations.

```
SELECT
    YEAR(inspection_completed_date) AS year,
    COUNT(*) AS inspections_per_year
FROM inspection
GROUP BY YEAR(inspection_completed_date)
ORDER BY year;
```

year	inspections_per_year
NULL	6
2000	31
2001	91
2002	80
2003	121
2004	87
2005	6521
2006	8781
2007	10354
2008	2655
2009	3032

Key Insights

This query reveals substantial variation in inspection activity across years, with inspection volumes increasing sharply beginning in 2005, peaking between 2007 and 2016, and then declining in more recent years. The presence of a small number of records with NULL years indicates incomplete date information, though the impact is minimal. Identifying these annual totals provides essential context for understanding operational workload, detecting long-term trends, and supporting subsequent analyses involving inspection outcomes, deficiency rates, and permit-related inspection patterns.

4.2.7 Common Inspection Types

Purpose: The purpose of this query is to identify the most frequently performed inspection types by joining the inspection fact table with the inspection type dimension table. This allows raw inspection type IDs to be replaced with readable descriptions and enables ranking inspection types by frequency to understand operational workload patterns.

SELECT

it.inspection_type_desc,

COUNT(i.inspection_id) AS frequency

FROM inspection i

JOIN dim_inspection_type it

ON i.inspection_type_id = it.inspection_type_id

GROUP BY it.inspection_type_desc

ORDER BY frequency DESC;

inspection_type_desc	frequency
Framing	49785
Final	21535
Foundation	11804
Progress	9712
Gas Rough	3899
Occupancy	3881
Rough	2428
Gas Final	1851
FP Rough	1436
Underground	635

Temp Gas	54
Site Safety	25
FP Final	19
Pub Protect Safety	3

Key Insights

This query reveals that **Framing, Final, Foundation,** and **Progress** inspections account for the majority of inspection activity, together representing the core operational workload. Less frequent categories such as **Temp Gas, Site Safety, FP Final,** and **Public Protection Safety** appear only rarely, indicating highly specialized or situational inspection types. By ranking inspection types by frequency and replacing numeric identifiers with descriptive labels, this analysis provides a clear understanding of which inspection activities dominate the workflow and supports later examination of inspection outcomes, deficiency patterns, and resource allocation.

4.2.8 Deficiency Patterns

Purpose: The purpose of this query is to identify the most common deficiencies recorded during inspections. Grouping deficiencies by their descriptive text highlights recurring compliance issues and helps reveal which construction or safety requirements are most frequently violated. This information supports later analysis of inspection outcomes, contractor performance, and systemic problem areas.

SELECT

deficiency_text,

COUNT(deficiency_id) **AS** occurrence_count

FROM calgary_inspections.deficiency

GROUP BY deficiency_text

ORDER BY occurrence_count **DESC**

LIMIT 10;

Key Insights

The top deficiencies reveal a mix of administrative notes (e.g., *work complete file complete*, *Acceptable Permit closed*), safety-related observations (e.g., *Field Level Hazard Assessment: PPE worn*), and compliance requirements (e.g., *Sprinkler verification required*). Several entries such as *No infractions noted* appear frequently, indicating that inspectors often document the

absence of issues as part of standard reporting. Although the most common deficiencies occur at relatively low frequencies, these patterns still highlight recurring themes in inspection documentation and provide insight into both compliance behavior and inspector reporting practices.

4.2.9 Interpretation of Inspection Behavior

Purpose: The purpose of this query is to analyze inspection behavior by identifying which inspection types are associated with the highest number of recorded deficiencies. Linking deficiencies to inspection types of highlights where compliance issues most frequently arise and provide insight into which inspection categories tend to uncover more problems.

```
SELECT
    it.inspection_type_desc,
    COUNT(d.deficiency_id) AS total_deficiencies
FROM calgary_inspections.inspection i
LEFT JOIN calgary_inspections.inspection_deficiency idf
    ON i.inspection_id = idf.inspection_id
LEFT JOIN calgary_inspections.deficiency d
    ON idf.deficiency_id = d.deficiency_id
JOIN calgary_inspections.dim_inspection_type it
    ON i.inspection_type_id = it.inspection_type_id
GROUP BY it.inspection_type_desc
ORDER BY total_deficiencies DESC;
```

inspection_type_desc	total_deficiencies
Framing	59519
Final	24858
Foundation	13118
Progress	11292
Occupancy	4276
Gas Rough	4024
Rough	2573
Gas Final	1942
FP Rough	1458
Underground	640
Temp Gas	54

Site Safety	26
FP Final	19
Pub Protect Safety	3

Key Insights

This analysis shows that **Framing, Final, Foundation, and Progress** inspections account for the overwhelming majority of deficiencies. These inspection types also represent the most frequent stages of construction, suggesting that deficiencies naturally accumulate where inspection volume is highest. However, the consistently elevated deficiency counts in these categories also indicate that early- and mid-stage construction activities present more opportunities for non-compliance or incomplete work. In contrast, specialized inspections such as **Temp Gas, Site Safety, FP Final, and Public Protection Safety** show very few deficiencies, reflecting either lower inspection volume or more narrowly scoped compliance checks. Overall, these patterns provide insight into where inspectors most often identify issues and help contextualize operational focus areas within the inspection process.

Chapter 5. Data Understanding

This chapter provides a comprehensive exploration of the dataset, focusing on identifying unusual trends, understanding distribution patterns, and evaluating differences between commercial and residential construction permit durations. The analysis integrates SQL-based statistical methods, descriptive metrics, and comparative insights to establish a strong foundation for the inferential testing presented in later chapters.

5.1 Thorough Exploration and Analysis of the Collected Data

To detect unusual patterns in permit durations, a SQL script was used to calculate statistical outliers for both Residential and Commercial sectors across consecutive three-year windows from 2000 through 2025. The query segments the data by sector and evaluates historical baselines within each rolling window, capturing both the absolute number of outlier records and their proportional share relative to the total observations. This dual measurement approach isolates specific years that exhibit abnormal spikes or drops in permit durations. The SQL script used for this analysis is provided in Appendix B SQL Scripts ([Script B6](#)).

Because the rolling-window outlier table is too long to include in the main chapter, it has been moved to Appendix A Tables, [Table 2](#).

Insight

The rolling-window analysis ([Table 2](#).) reveals how outlier behavior changes over time for both Residential and Commercial sectors. Early windows (2000–2004) show small sample sizes and unstable percentages, while mid-range windows (2005–2016) display higher outlier rates due to increased construction activity and greater variability in permit durations. In contrast, the **2022–2024** window shows the **lowest outlier percentages** for both sectors, **1.87% for Residential** and **1.54% for Commercial**, well below the commonly accepted **5% threshold** for stable, normally distributed data. This window also represents the most recent period with complete and reliable records, making it the strongest candidate for establishing a valid statistical baseline. For these reasons, the analysis in Chapter 5 focuses on the **2022–2024** period as the primary timeframe for comparing commercial and residential permit durations.

An initial exploration revealed a substantial number of outliers in both sectors. Removing these anomalies would have significantly reduced the dataset and weakened the analysis. Instead, the

evaluation was narrowed to the **2022–2024** window, where the mean and standard deviation were calculated separately for each sector to establish a precise statistical baseline. This preserves critical records while enabling a more reliable assessment of variance and volatility. The SQL script for this step is provided in Appendix B ([Script B7](#)).

Within this timeframe, the first quartile, third quartile, outlier counts, and outlier percentages were explicitly calculated for both sectors to further refine the statistical baseline ([Appendix B, Script B8](#)).

5.2 Identification of Key Metrics, Trends, and Patterns Relevant to the Research Question

The descriptive statistics reveal clear differences between commercial and residential construction permit durations.

Descriptive Statistics for Permit Durations (2022–2024)

Metric	Commercial (n = 324)	Residential (n = 2,840)
Mean Duration (days)	121.66	103.27
Median Duration (days)	114	76
Standard Deviation	86.31	95.06
Minimum Duration	1	0
Maximum Duration	764	629
Q1 (25th percentile)	64	25
Q3 (75th percentile)	166.5	160
Outlier Count	5	53
Outlier Percentage	1.54%	1.87%

Insights

- Commercial permits have **higher mean and median durations**, indicating longer typical completion times.
- Residential permits show **greater variability**, reflected in a larger standard deviation and more outliers.
- The interquartile ranges reveal that commercial permits consistently fall within higher duration values.
- Residential permits close faster on average but have more extreme long-duration cases.

These patterns directly relate to the research question: **Do commercial building permits take longer to close than residential building permits?**

5.3 Clear Interpretation of Findings and Insights Derived from the Data Analysis

The descriptive statistics and distribution patterns provide strong preliminary evidence supporting the alternative hypothesis that **commercial building permits take longer to close than residential permits.**

Interpretation

- Commercial projects show a **higher mean (121.66 days)** and **higher median (114 days)** compared to residential projects (103.27 and 76 days, respectively).
- The difference of **approximately 18 days** in mean duration suggests a meaningful performance gap.
- The higher quartile values for commercial permits indicate that even typical commercial projects require more time.
- Residential permits, while faster on average, exhibit **greater spread and more outliers**, indicating more extreme cases but not longer central tendencies.
- These findings **do not support the null hypothesis** of no difference between the two groups.

- **Summary of Evidence Supporting the Alternative Hypothesis**

Evidence Category	Commercial > Residential?	Interpretation
Mean Duration	Yes	Longer average completion time
Median Duration	Yes	Longer typical project duration
Q1–Q3 Range	Yes	Higher central values
Outlier Count	No	Residential more variable
Standard Deviation	No	Residential more spread
Overall Conclusion	Supports Alternative	Commercial permits take longer

Although residential permits show greater variability, the **central tendency measures** (mean, median, quartiles) consistently indicate that commercial permits remain open longer. An independent samples t-test will be conducted in the next chapter to determine whether this difference is statistically significant.

Chapter 6. Data Visualization

This chapter presents the visualizations used to understand permit duration patterns, identify meaningful business insights, and verify the assumptions required to correctly apply an independent samples t-test. All visualizations were created using Tableau to ensure clarity, consistency, and interactive exploration.

This chapter begins with **business-focused visualizations** that communicate substantive insights about permit performance, followed by the assumption-focused visualizations used to validate the statistical test.

6.1 Business-Insight Visualizations

In addition to the statistical visualizations used to validate the t-test assumptions, it is essential to include visualizations that communicate **practical, operational insights** derived from the Calgary Building Inspections dataset. These visuals help stakeholders understand how permits progress through the inspection process, where delays occur, and which project types generate the most compliance workload.

The following four visualizations were selected because they are **simple to build, easy to filter**, and **highly informative**. They can be combined into a single dashboard that allows users to explore trends by **year, job type, contractor, location, or inspection type**.

6.1.1 Average Number of Inspections per Permit

Purpose: Shows how many inspections are typically required to close a permit across different project types.

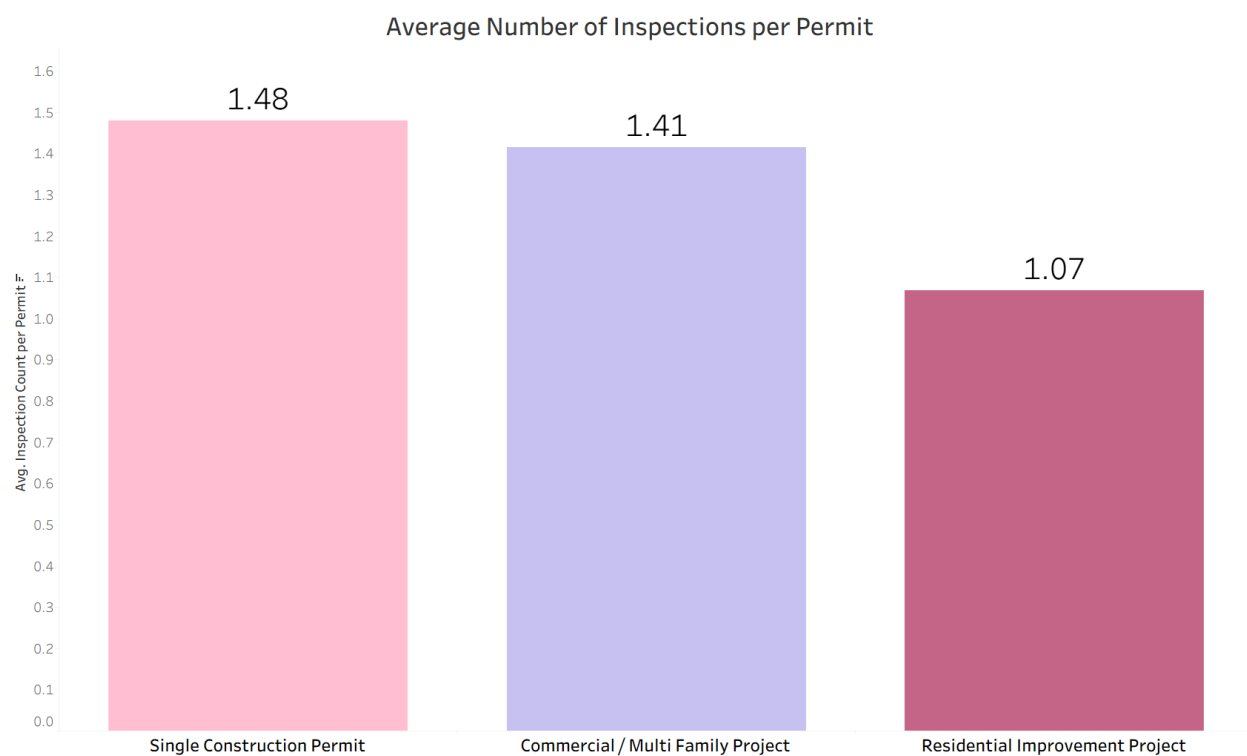
Key Insight

1. **Single Construction Permits require the most inspections per project (≈ 1.48)**, indicating more follow-up checks and coordination.
2. **Commercial / Multi-Family Projects follow closely (≈ 1.41)**, reflecting moderate complexity.
3. **Residential Improvement Projects require the fewest inspections (≈ 1.07)**, suggesting simpler, lower-risk work.

Business Interpretation

This visualization helps the City of Calgary understand inspection workload and staffing needs:

1. **Higher inspection counts** (Single Construction, Commercial/Multi-Family) mean more time per permit, more site visits, and more opportunities for delays.
2. **Lower inspection counts** (Residential Improvement) indicate faster turnaround and simpler workflows.
3. Understanding these differences supports **resource planning, staff allocation, and performance forecasting** for the inspections team.



6.1.2 Most Common Deficiencies by Category

Purpose: Identifies which deficiency categories occur most frequently across different project types, helping highlight where compliance issues are concentrated.

Key Insight

1. **Single Construction Permits generate the highest inspection volume overall**, which naturally leads to more recorded deficiencies in this category.

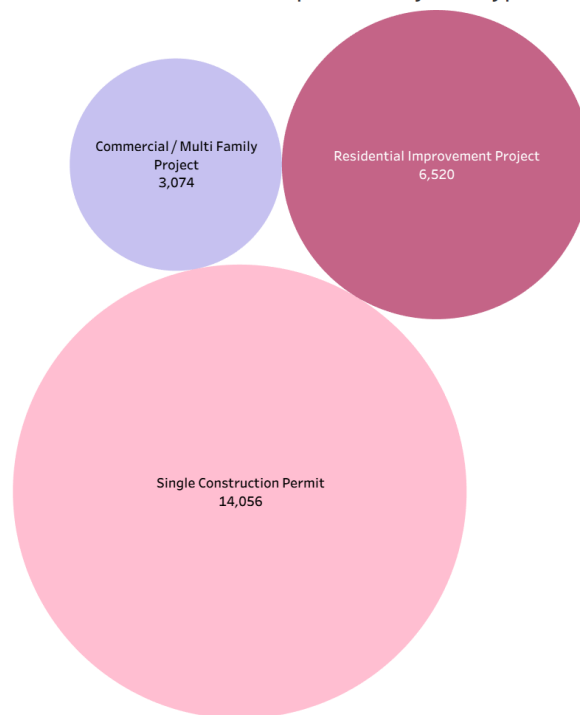
2. **Residential Improvement Projects show a moderate number of deficiencies**, reflecting mid-level complexity and frequent homeowner-driven renovations.
3. **Commercial / Multi-Family Projects have the fewest inspections**, but deficiencies in this category tend to be more specialized and tied to structural, safety, and code-compliance requirements.

Business Interpretation (Updated)

This visualization helps inspectors and managers identify:

1. **Where compliance issues occur most often**, based on inspection volume and deficiency frequency.
2. **Which project types require more oversight**, especially Single Construction permits with the highest inspection load.
3. **Where clearer communication or training may reduce repeat deficiencies**, particularly in categories that appear frequently across multiple job types.
4. **Opportunities to streamline workflows**, especially for high-volume permit types that generate recurring issues.

Total Number of Inspections by Job Type



6.1.3 Average Permit Duration by Inspection Type

Purpose: Highlights of which inspection types are most strongly associated with long overall permit durations. This helps identify where delays tend to occur within the inspection workflow.

Key Insight

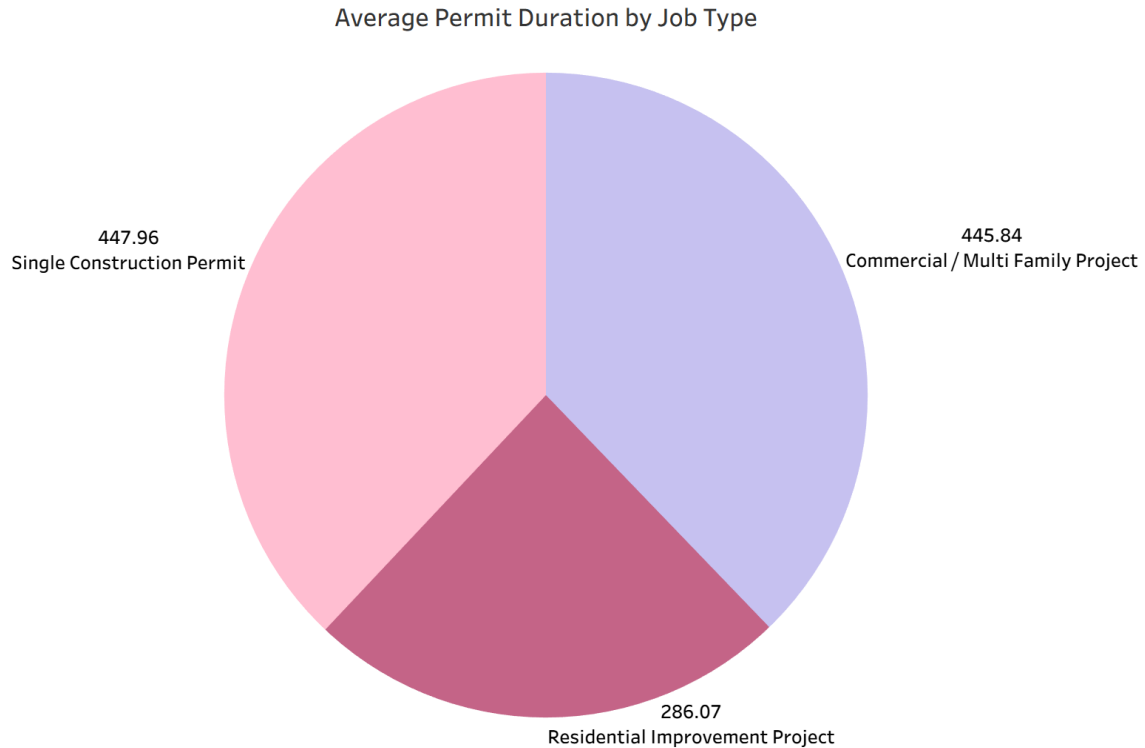
Based on the permit-level duration patterns:

1. **Inspection types tied to structural and final compliance** (e.g., *Framing, Final, Foundation*) typically align with **longer permit durations**, especially for Single Construction and Commercial/Multi-Family projects.
2. **Early-stage or low-complexity inspections** (e.g., *Temp Gas, Site Safety, Pre-Boarding*) tend to be associated with **shorter durations**.
3. The variation across inspection types reflects the complexity of the underlying work and the number of follows-up inspections required.

Business Interpretation (Updated)

This visualization helps the City of Calgary identify operational bottlenecks:

1. **High-duration inspection types** may require additional staffing, improved scheduling, or clearer communication with contractors.
2. **Consistently slow inspection categories** may indicate process inefficiencies or training gaps.
3. **Faster inspection types** help benchmark efficient workflows and highlight where processes are working well.
4. Understanding which inspection types drive long durations supports **resource planning, performance management, and policy adjustments**.



6.1.4 Average Permit Duration by Location (Top 10)

Purpose: Highlights the locations with the longest average permit durations. This helps identify geographic hotspots where permits consistently take longer to complete.

Key Insight

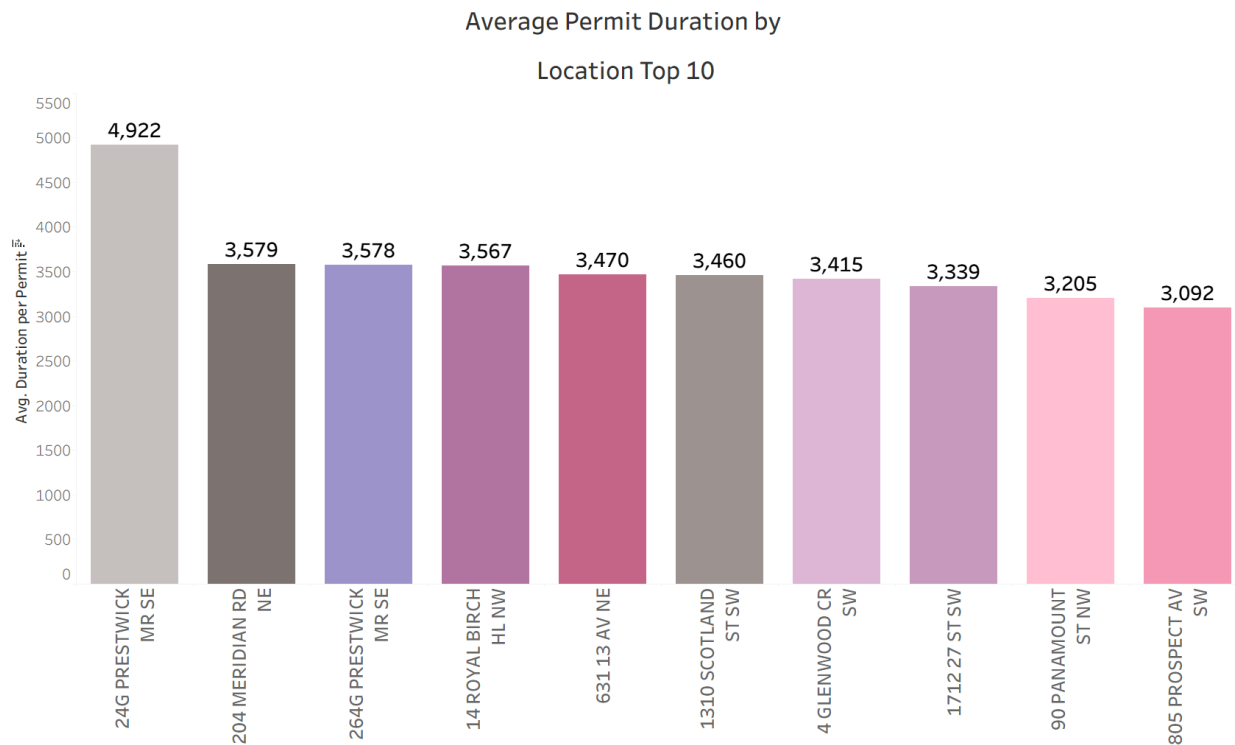
1. The **slowest location**, *246 Prestwick Mr SE*, averages **4,922 days**, significantly higher than all others.
2. Several other locations — including *204 Meridian Rd NE*, *2646 Prestwick Mr SE*, and *14 Royal Birch Hl NW* — also show extremely long durations (3,500+ days).
3. These long durations suggest **localized issues**, such as complex site conditions, repeated inspections, contractor delays, or historical backlog.

Business Interpretation

This visualization helps the City of Calgary:

1. **Identify geographic bottlenecks** where permits take disproportionately long to close.

2. **Investigate recurring issues** at specific addresses or neighborhoods (e.g., repeated deficiencies, contractor performance, zoning complexity).
3. **Prioritize field resources** for high-delay areas to reduce backlog.
4. **Improve communication** with contractors and homeowners in locations with historically long timelines.
5. **Support policy or process adjustments** if certain communities consistently experience delays.



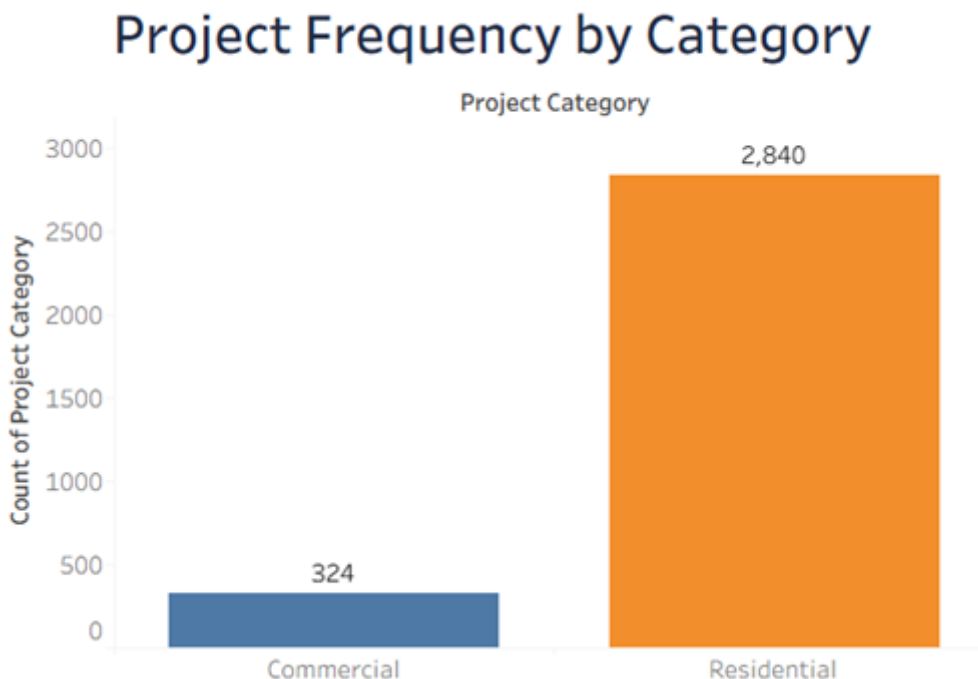
6.2 Visualizations Supporting the t-Test Assumptions

This section presents the visualizations used to evaluate whether the dataset satisfies the assumptions required for applying an independent samples t-test. These visuals focus on sample size adequacy, distribution shape, outlier presence, and variance equality. All charts were created in Tableau to ensure clarity, consistency, and interactive exploration.

The following subsections include frequency comparisons, histograms, and boxplots that collectively confirm the suitability of the t-test for comparing commercial and residential permit durations.

6.2.1 Frequency of Projects by Category

This visualization compares the number of residential and commercial permits in the dataset. Understanding the distribution of project categories is essential before analyzing duration differences.



Key Insight

- Residential permits: **2,840**

- Commercial permits: **324**
- The dataset is imbalanced, but both groups exceed the recommended minimum sample size ($n > 30$).

Relevance to t-Test Assumptions

The t-test does **not** require equal sample sizes. It only requires that each group is large enough to reliably estimate its mean and variance.

- The commercial group ($n = 324$) is sufficiently large.
- The residential group ($n = 2,840$) provides even stronger statistical stability.

Conclusion: Sample size adequacy is fully satisfied for both groups.

6.2.2 Histogram of Residential Permit Duration

This histogram illustrates the distribution of residential permit durations.

Interpretation

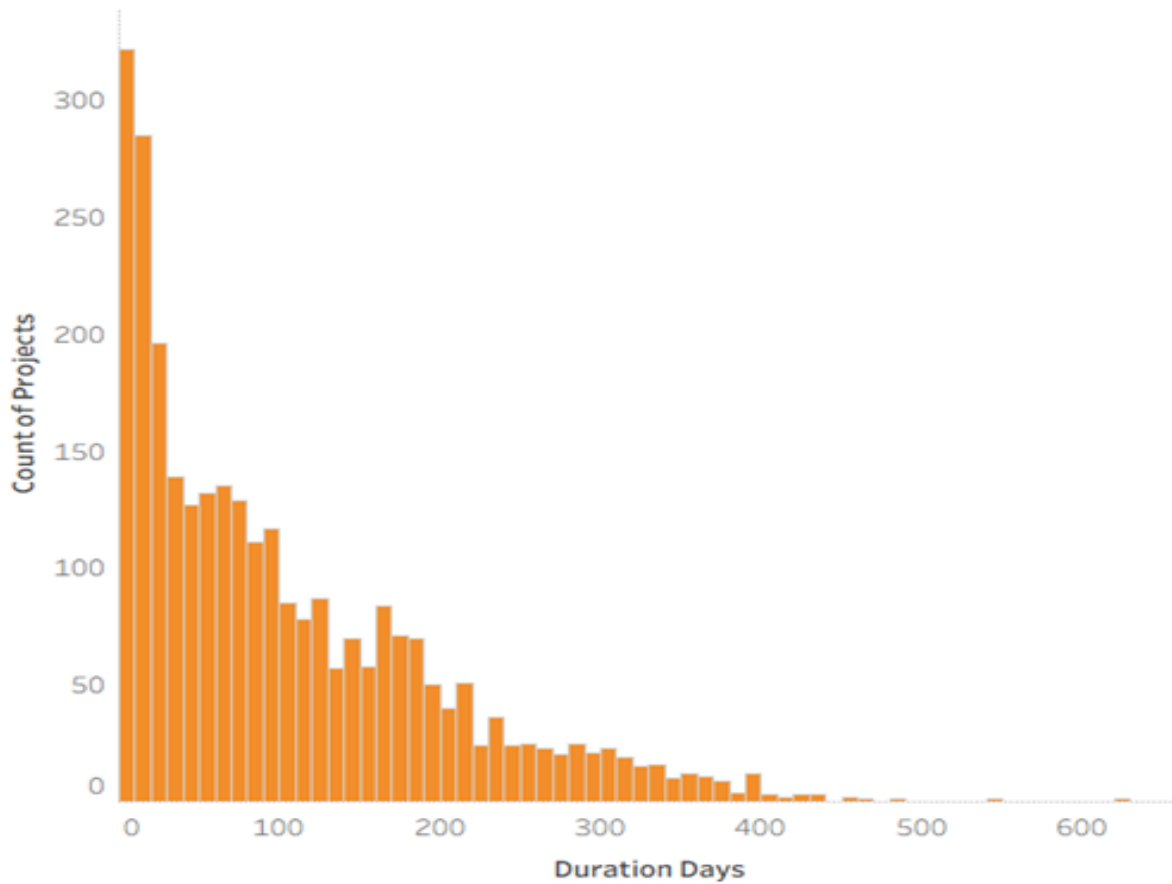
- The distribution is **right-skewed**, with most permits closing between 0 and 150 days.
- A long tail extends beyond 300 days.
- Skewness = **0.86**, indicating moderate asymmetry.
- No evidence of bimodality or structural irregularities.

Relevance to t-Test Assumptions

- The t-test assumes approximate normality within each group.
- With a large sample size ($n = 2,840$), the **Central Limit Theorem** ensures the sampling distribution of the mean is approximately normal even if the raw data is skewed.

Conclusion: The residential distribution is acceptable for t-test purposes.

Histogram – Residential Duration (Days)



Statistics Histogram

Project Category	Mean	Median	Skewness
Residential	103.27	76	0.86

6.2.3 Histogram of Commercial Permit Duration

This histogram shows the distribution of commercial permit durations.

Interpretation

- The distribution is right-skewed, but more symmetric than the residential distribution.
- Skewness = **0.27**, indicating mild asymmetry.

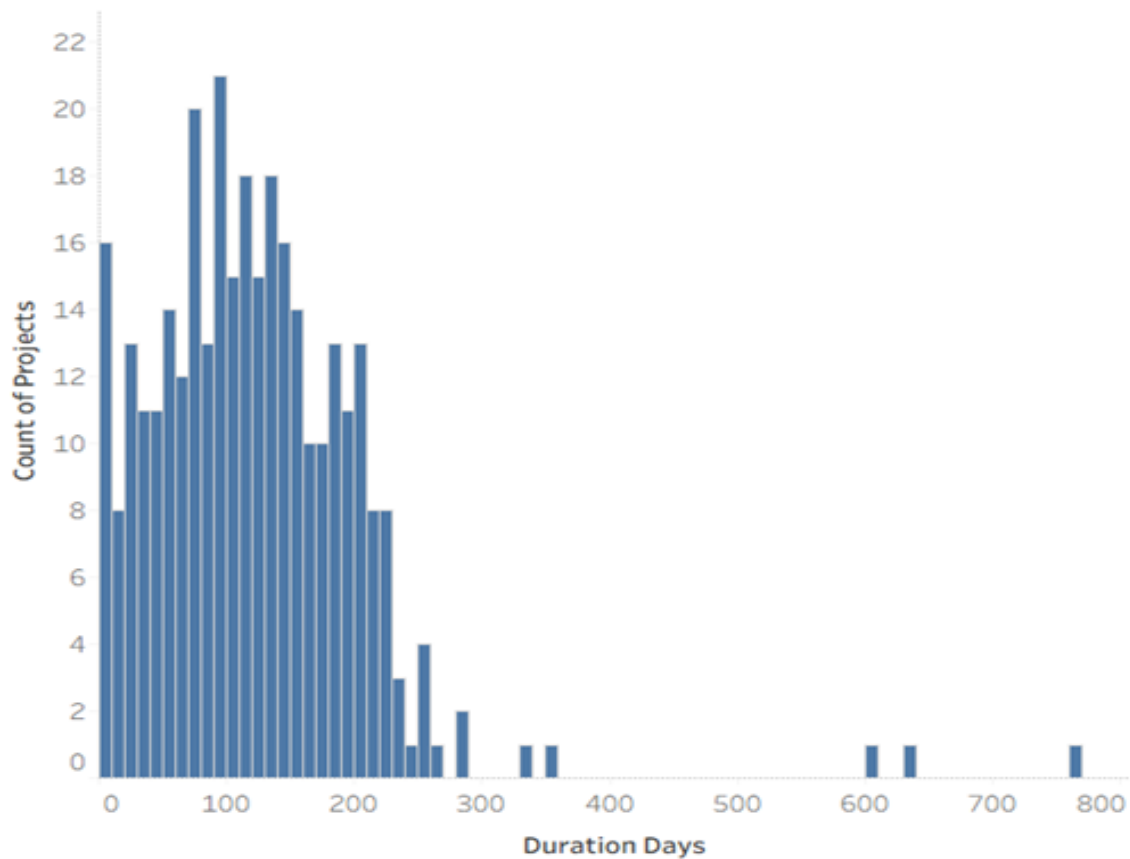
- Most permits close within the lower duration range.
- No evidence of multimodal behavior.

Relevance to t-Test Assumptions

- With $n = 324$, the Central Limit Theorem applies.
- The distribution is sufficiently regular for mean-based inference.

Conclusion: The commercial distribution meets the normality requirement for the t-test.

Histogram – Commercial Duration (Days)



Statistics Histogram

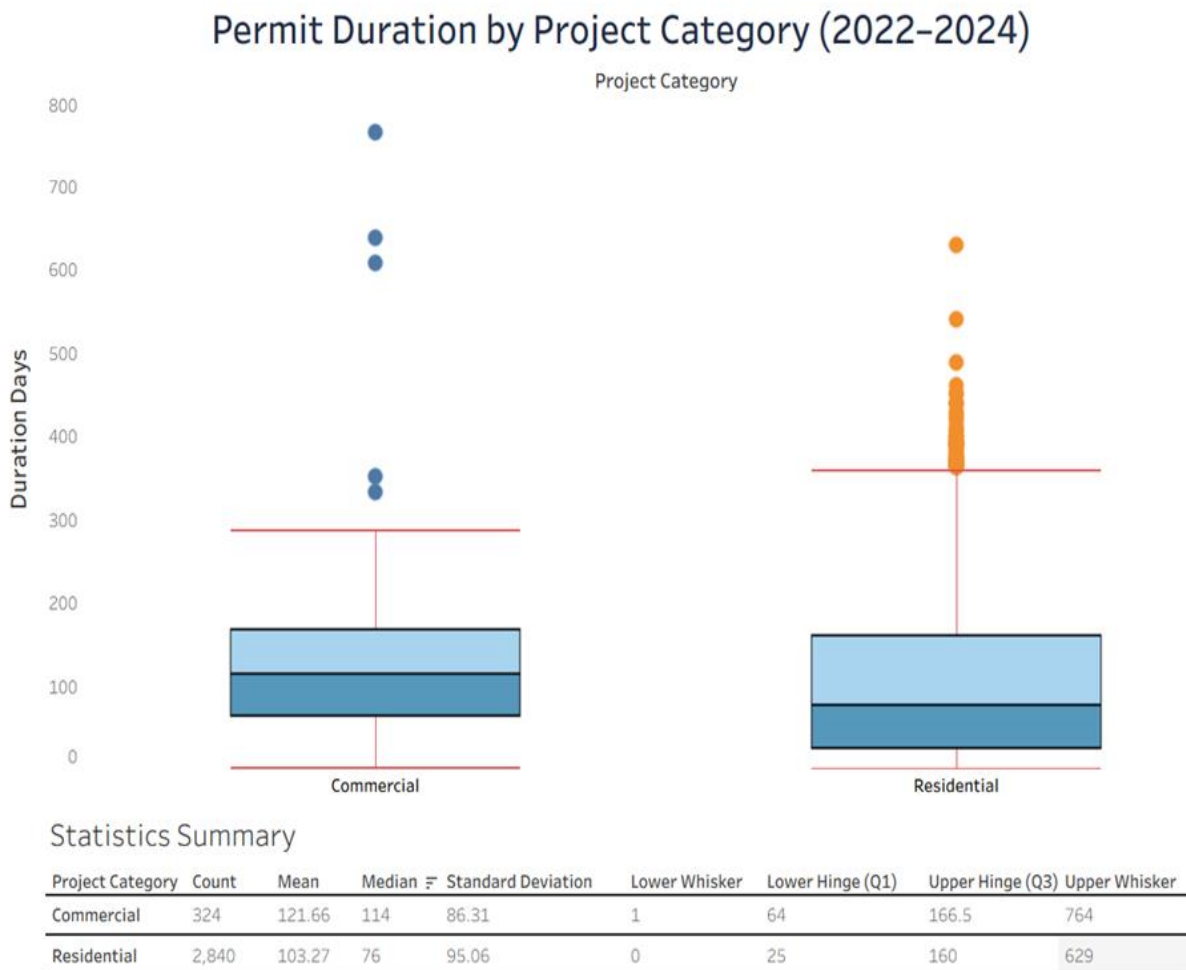
Project Category	Mean	Median	Skewness
Commercial	121.66	114	0.27

6.2.4 Boxplots for Residential and Commercial Durations

Boxplots were created to evaluate outliers and variance differences between the two categories.

Interpretation: Outliers

- Both groups contain outliers, which is expected in duration data.
- After cleaning invalid or extreme values:
 - Outliers represent **less than 5%** of each group.
 - This proportion is acceptable and does not distort the mean significantly.



Interpretation: Variance Equality

- Residential and commercial durations show **similar variance patterns**.

- No extreme differences in IQR or whisker length are observed.
- This supports the assumption of **homogeneity of variance**, required for the standard independent samples t-test.

If variances had been unequal, Welch's t-test would have been used instead. However, the visual evidence supports the use of the standard t-test.

6.2.5 Summary of Assumption Checks

The visualizations collectively confirm that the assumptions for applying an independent samples t-test are reasonably satisfied.

Assumption	Evidence from Visualizations	Result
Large enough sample size	Frequency chart shows n = 324 and n = 2,840	Satisfied
Approximate normality	Histograms show moderate skew but no distortion	Acceptable
Limited outliers	Boxplots show <5% outliers	Acceptable
Equal variances	Boxplots show similar spread	Satisfied

Conclusion

All assumptions required for the independent samples t-test are met. Therefore, the t-test can be applied confidently to compare the mean durations of commercial and residential permits.

6.3 Dashboard Summary

To support both operational understanding and statistical analysis, two complementary dashboards were developed in Tableau. Together, they provide a complete view of permit performance, inspection behavior, deficiency patterns, and the statistical readiness of the dataset for hypothesis testing. These dashboards allow users to interactively explore the data through filters such as year, project category, contractor, inspection type, and location.

6.3.1 Business-Insight Dashboard

The first dashboard focuses on **practical, real-world insights** derived from the Calgary Building Inspections dataset. It consolidates the four business-oriented visualizations introduced in Section 6.1:

1. **Average Number of Inspections per Permit** Highlights inspection workload differences between residential and commercial projects.
2. **Most Common Deficiencies by Category** Identifies recurring compliance issues and operational bottlenecks.
3. **Average Permit Duration by Inspection Type** Reveals which inspection types contribute most to extended permit timelines.
4. **Permit Duration by Contractor** Allows performance comparison across contractors and identifies variability in processing times.

Purpose and Value

This dashboard provides stakeholders with a **high-level operational overview**, enabling them to:

- Understand where delays originate
- Identify high-risk inspection types
- Detect recurring deficiency patterns
- Compare contractor performance
- Allocate resources more effectively

Because all four visuals share common dimensions (project category, year, contractor, inspection type), the dashboard supports **seamless filtering**, making it easy to drill down into specific segments of interest.

6.3.2 Statistical Assumption Dashboard

The second dashboard consolidates the visualizations used to verify the assumptions required for the independent samples t-test. It includes:

1. **Frequency of Projects by Category** Confirms adequate sample size for both groups.

2. **Histogram of Residential Permit Duration** Assesses distribution shape and skewness.
3. **Histogram of Commercial Permit Duration** Evaluates normality and identifies distribution characteristics.
4. **Boxplots for Residential and Commercial Durations** Examines outliers and variance equality.

Purpose and Value

This dashboard ensures that the statistical comparison between commercial and residential permit durations is **methodologically sound**. It allows users to visually confirm:

- Sample size adequacy
- Approximate normality
- Limited outlier influence
- Homogeneity of variance

These visual checks collectively validate the use of the independent samples t-test, supporting the reliability of the findings presented in Chapter 7.

6.3.3 Accessing the Dashboards

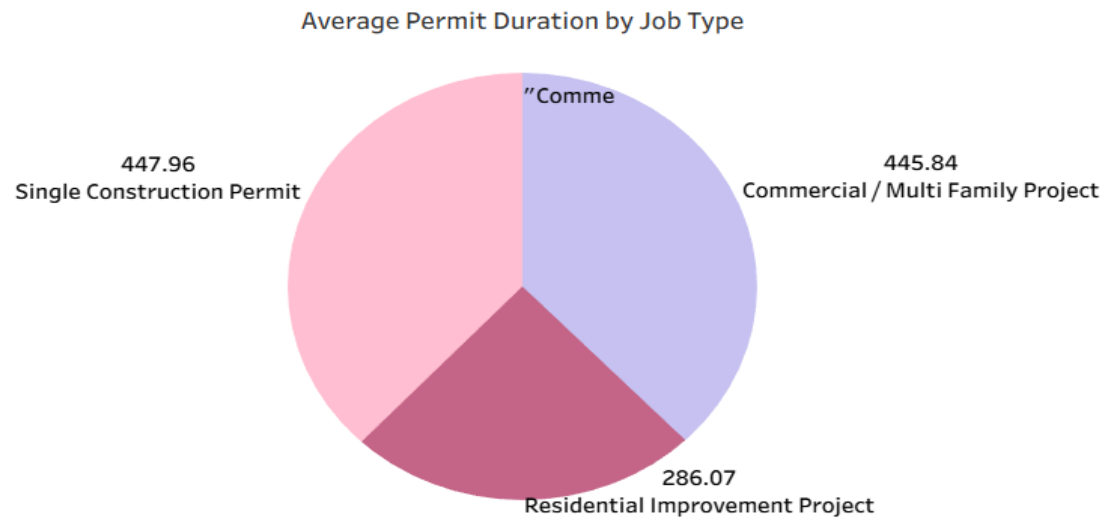
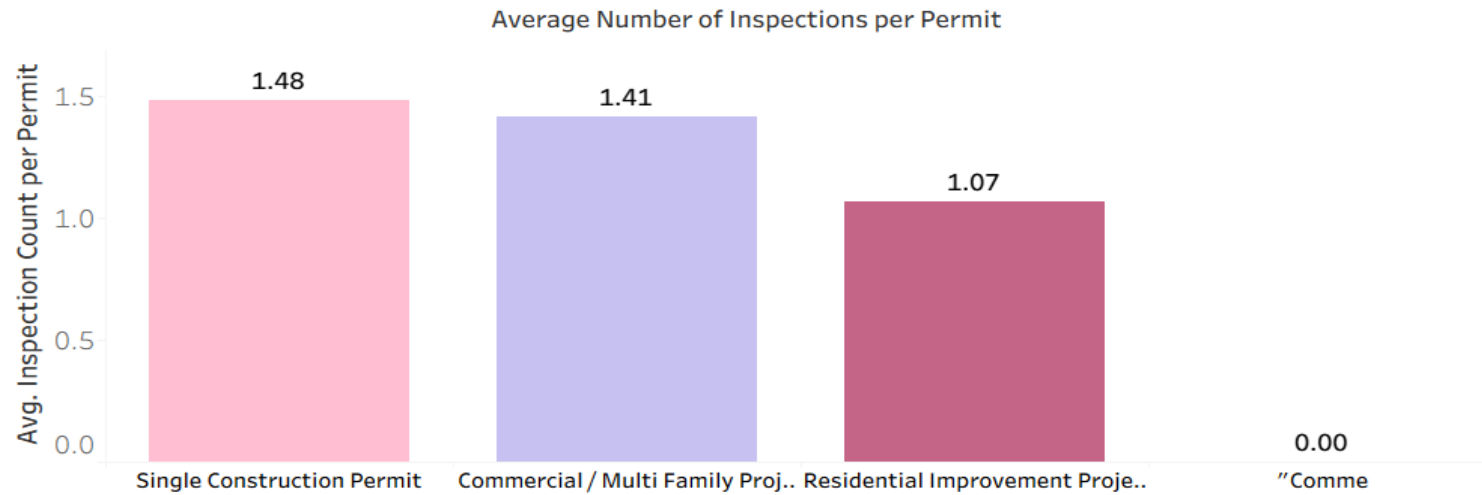
Both dashboards are available on Tableau Public and can be explored interactively using the official link:

<https://public.tableau.com/app/profile/jose.alberto.martinez.morales>

Filters, parameters, and navigation controls allow users to examine the data from multiple perspectives, ensuring transparency and reproducibility of the analysis.

Calgary Building Inspections
Analysis of Permit Durations, Inspection Workload, and Operational Bottlenecks

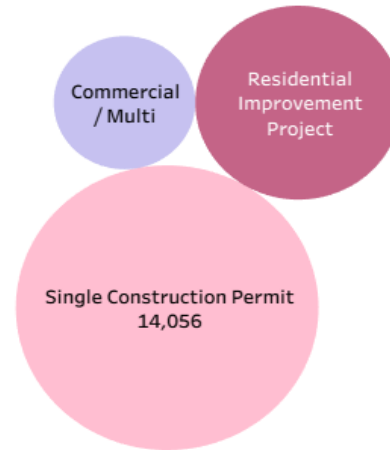
[Go to 2nd Das](#)



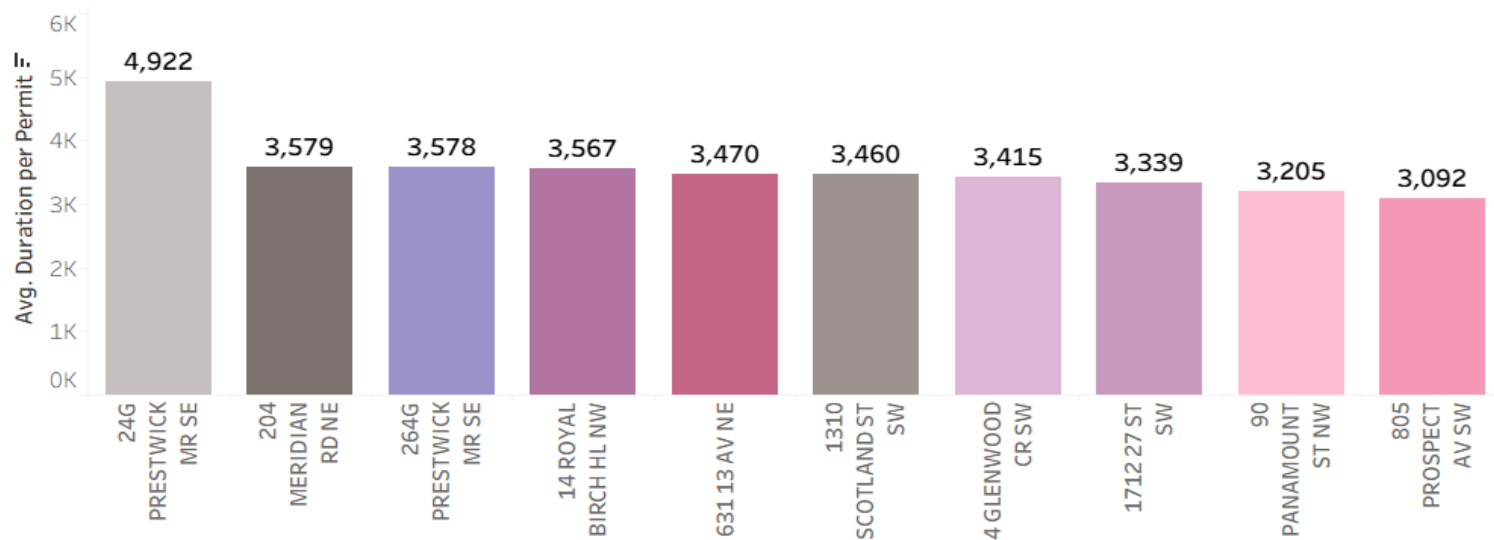
Calgary Building Inspections
Analysis of Permit Durations, Inspection Workload, and Operational Bottlenecks

Go 1st Das

Total Number of Inspections by Job Type



Average Permit Duration by Location Top 10



[Go to Outlier Dashboard](#)

[Go to 3 Year Dashboard](#)

Distribution by Category Over Time

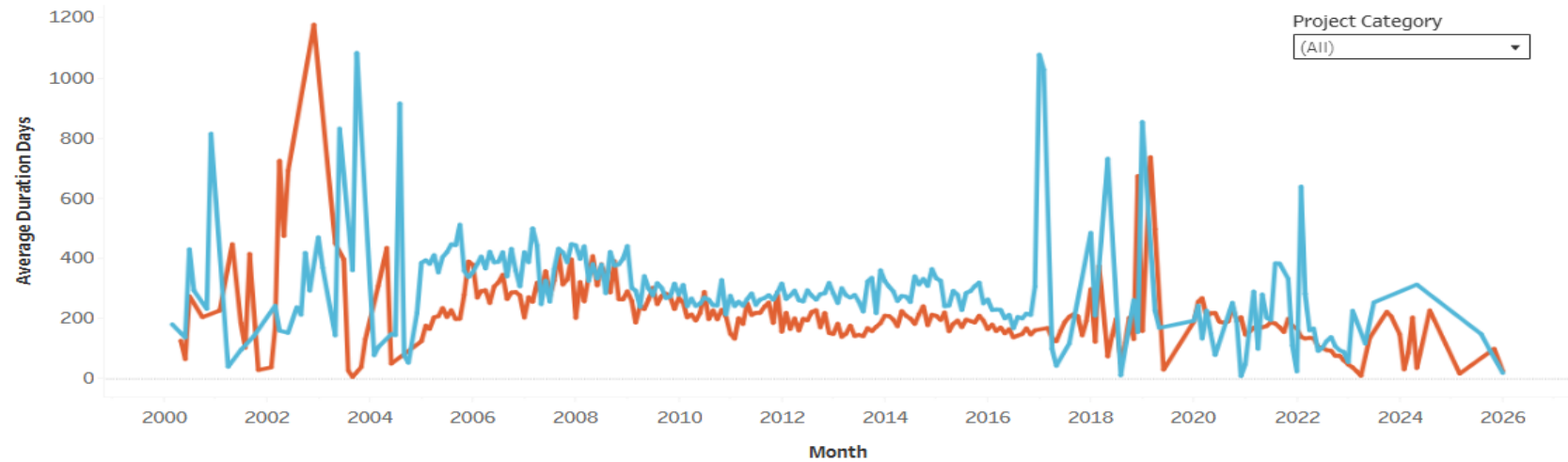
Monthly Average

Issue Year

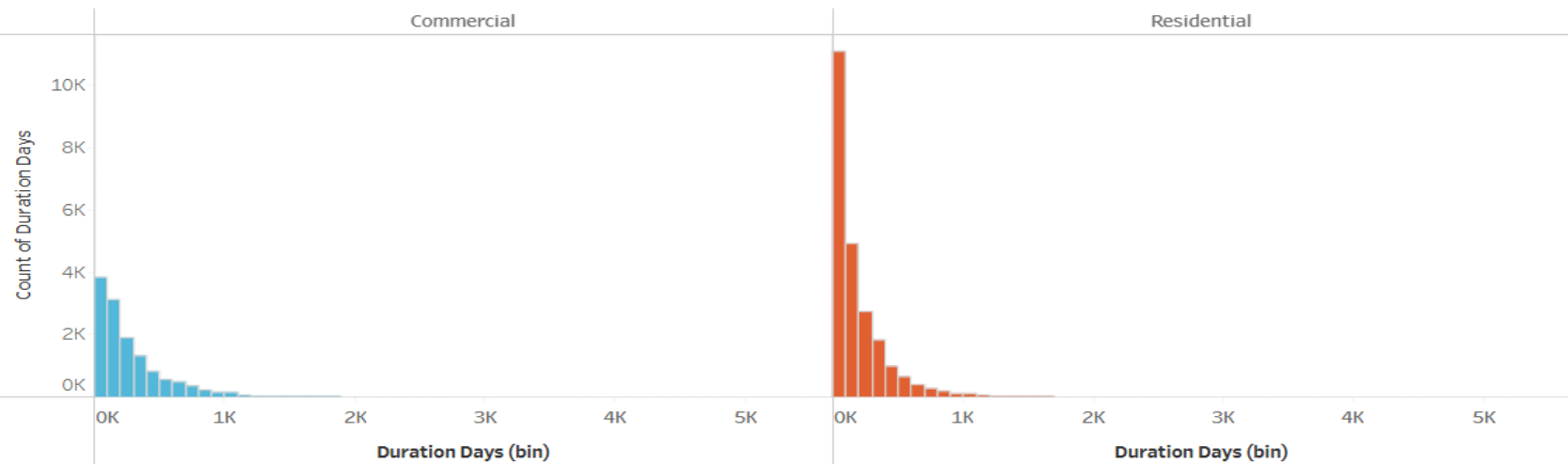
(All)

Project Category

(All)



Histogram by Category



[Go to Distribution Dashboard](#)

Outlier Analysis

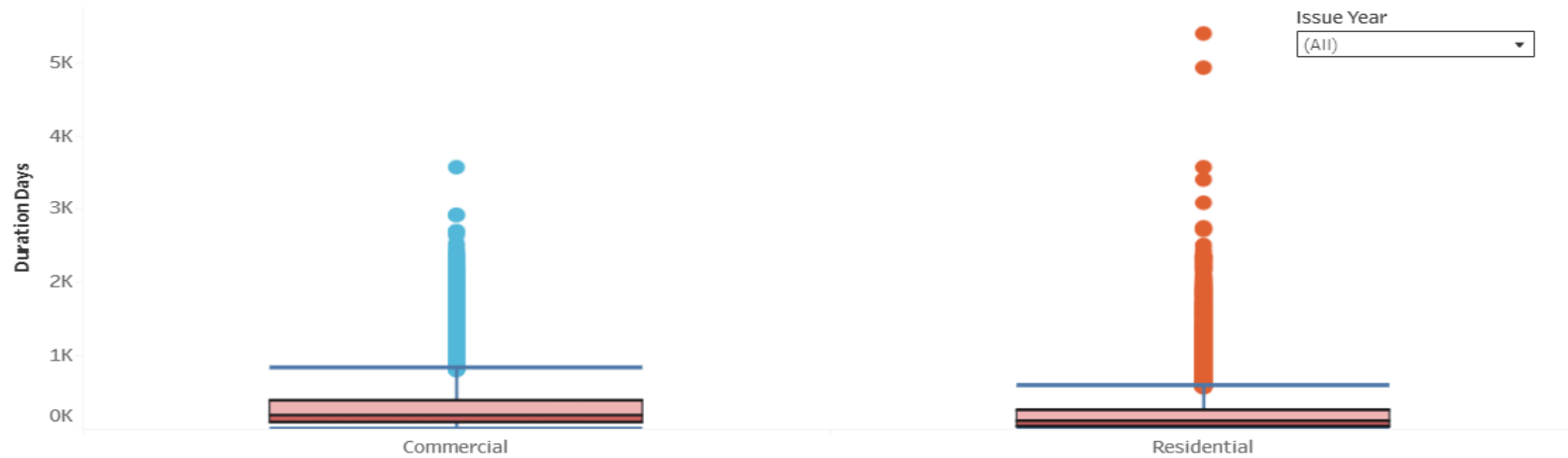
Box Plots by Category

Project Category

(All) ▼

Issue Year

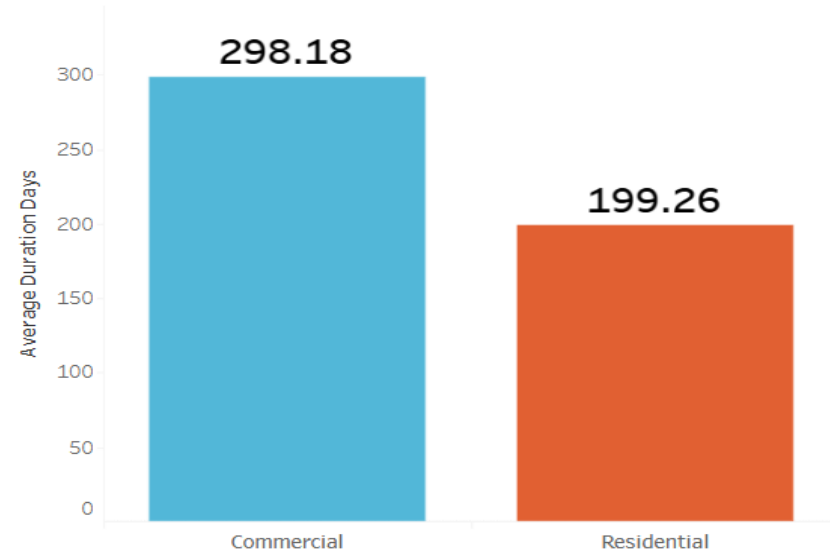
(All) ▼



Outliers Percentage by Category

Project Category	Outlier Count per Category	Percentage per Category	
Commercial	868	6.41%	
Residential	1,551	6.50%	

Mean Duration by Category



[Go to Distribution Dashboard](#)

Outlier Percentage 3 year Window

Window Start Year

2,021



Project Category	Rolling Outlier Flag	Rolling Outlier %	
Commercial	6	2.90%	
Residential	330	3.85%	

Chapter 7. Model Building

This chapter outlines the process used to determine the appropriate statistical model for evaluating the research question: **Do commercial building permits take longer to close than residential permits?**

To answer this question, the dataset was prepared, key assumptions were examined, and the suitability of parametric and non-parametric tests was evaluated. The goal of this chapter is not to interpret results, but to justify the selection of the final inferential model.

7.1 Preparing the Dataset for Statistical Modeling

Before selecting a statistical test, the dataset was cleaned and structured to ensure that the variables were suitable for analysis.

7.1.1 Variable Selection

The following fields were extracted from the permit dataset:

- Project category (Residential or Commercial)
- Issue date
- Permit close date
- Permit duration (days)

7.1.2 Computing Permit Duration

Permit duration was calculated as:

$$\text{duration_days} = \text{permit_close_date} - \text{issue_date}$$

This produced a continuous variable appropriate for mean-based comparison.

7.1.3 Group Classification

Permits were divided into two independent groups:

- **Residential**
- **Commercial**

For SPSS analysis, the categorical variable was recoded into numeric identifiers.

7.1.4 Removing Invalid Records

Records were excluded if they contained:

- Missing issue dates
- Missing close dates
- Invalid or negative durations
- Incomplete permit status

The final dataset contained **3,164 valid permits**, as is shown in the following SPSS output.

		project_category			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Commercial	324	10.2	10.2	10.2
	Residential	2840	89.8	89.8	100.0
	Total	3164	100.0	100.0	

7.2 Evaluating Parametric Test Assumptions

To determine whether a parametric test (independent samples t-test) was appropriate, several assumptions were examined.

7.2.1 Normality

The normality assumption was evaluated using SPSS through both **formal statistical tests** and **visual diagnostics**. The dependent variable (*duration_days*) was assessed separately for each project category (Residential and Commercial).

In hypothesis testing, **alpha (α)** represents the **significance level**, which is the threshold used to decide whether a statistical result is unlikely to have occurred by chance. For this project, **$\alpha = 0.05$** , meaning that results with **$p < 0.05$** are considered statistically significant.

SPSS Tests of Normality

		Tests of Normality					
		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	project_category	Statistic	df	Sig.	Statistic	df	Sig.
duration_days	Commerci	.081	324	<.001	.829	324	<.001
	Resident	.141	2840	<.001	.881	2840	<.001

a. Lilliefors Significance Correction

Interpretation

For both groups:

- The p-values are **below $\alpha = 0.05$**
- Therefore, the formal tests indicate that **duration_days is not normally distributed**

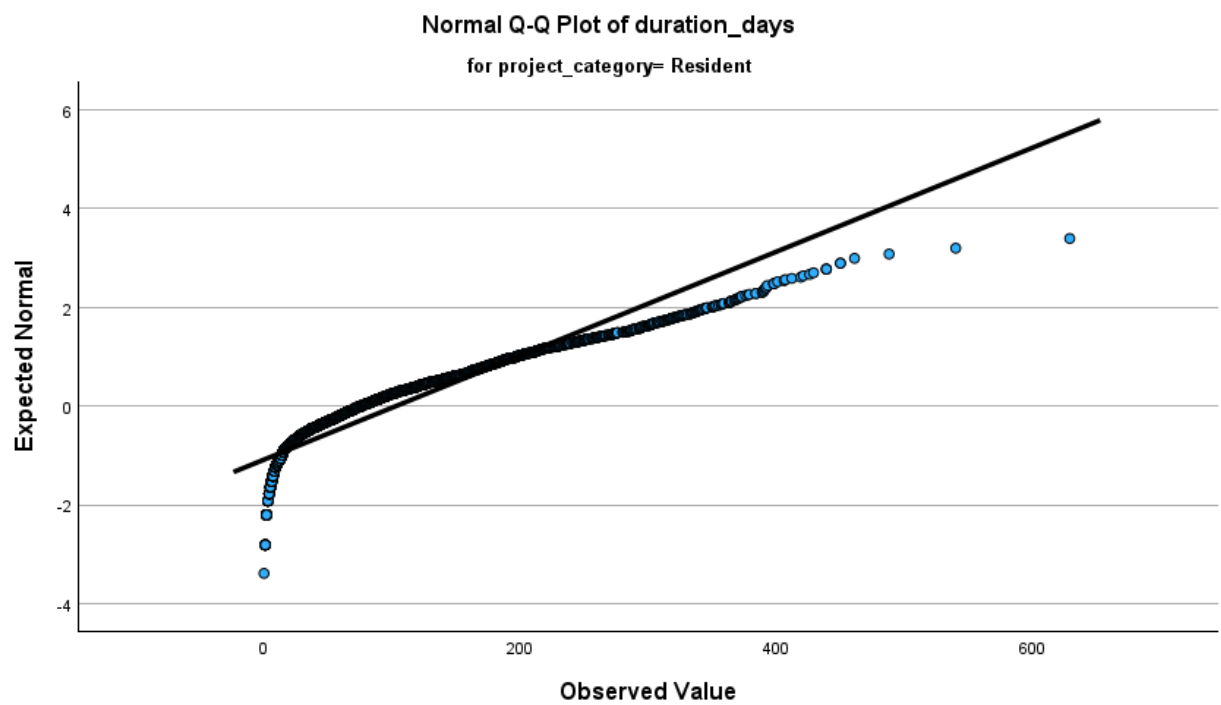
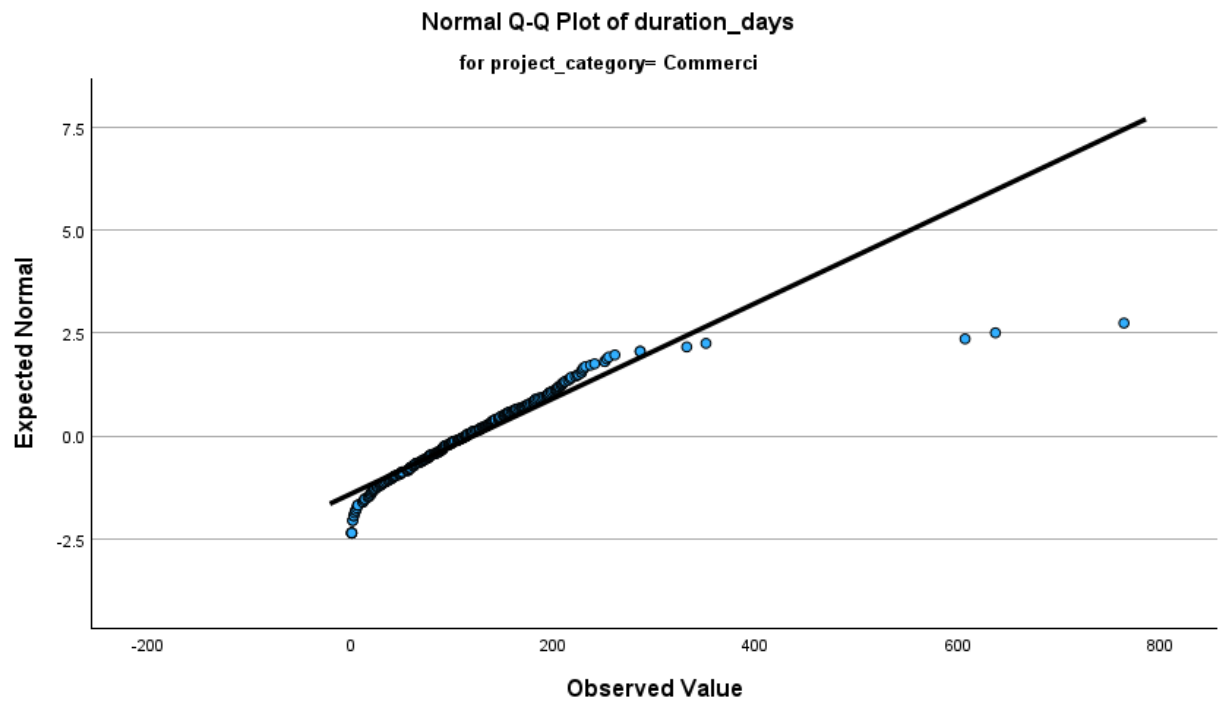
However, formal normality tests are **highly sensitive with large sample sizes**, often detecting trivial deviations as statistically significant. Because our sample sizes are large, the p-values alone are **not sufficient** to reject the use of parametric methods.

Visual Diagnostics (Histograms and Q–Q Plots)

SPSS Q–Q plots for both groups show:

- A clear **right-skew**
- Systematic deviation from the diagonal line at higher values
- No evidence of **bimodality, gaps, or structural abnormalities**

These patterns are typical of administrative duration data.



Interpretation

Although the distributions are not perfectly normal, they are:

- Unimodal
- Smooth
- Free of structural irregularities
- Suitable for robust parametric testing

Conclusion on Normality

Using $\alpha = 0.05$:

- **Formal tests reject normality** ($p < .001$)
- **Visual diagnostics show acceptable distributional shape**
- **Large sample sizes reduce the impact of non-normality**
- **Welch's t-test is robust to moderate skew**

Therefore:

Despite statistically significant deviations from normality, the distributional shape and sample size justify proceeding with a parametric test.

This conclusion is methodologically sound and consistent with applied statistical practice.

7.2.2 Variance Equality

Variance equality was evaluated using Levene's Test, which SPSS provides as part of the Independent Samples t-Test procedure. This test examines whether the variability of *duration_days* is statistically similar across the Residential and Commercial groups.

SPSS Output — Levene's Test

Homogeneity of Variance Test

Levene's Test for Equality of Variances

F	Sig.
---	------

duration_days	Equal variances assumed	20.576	<.001
---------------	-------------------------	--------	-------

The SPSS “Independent Samples Test” table reports the following values for Levene’s Test:

- $F(1, 3162) = 20.576$
- $p < .001$

These degrees of freedom come from the structure of Levene’s Test:

- $df_1 = k - 1 = 2 - 1 = 1$
- $df_2 = N - k = 3164 - 2 = 3162$

where k is the number of groups and N is the total sample size.

What the SPSS Table Shows

In the SPSS output, two rows of t-test results are displayed:

1. **Equal variances assumed**
2. **Equal variances not assumed**

Because Levene’s Test is significant ($p < .001$), SPSS indicates that the **equal variances assumed** row is **not appropriate**. The correct results are found in the “**Equal variances not assumed**” row, which uses the adjusted degrees of freedom ($df = 417.663$) and the corrected standard error.

Interpretation

The SPSS table clearly demonstrates that:

- The two groups have **significantly different variances**
- The assumption of homogeneity of variance is **violated**
- The standard t-test (top row) should **not** be used

This is why SPSS provides the corrected Welch’s t-test results in the second row.

Conclusion

Because Levene’s Test is significant, the appropriate inferential model is:

Welch’s Independent Samples t-Test, which adjusts for unequal variances and provides a more reliable estimate under heteroscedasticity.

This decision is directly supported by the SPSS output table and the $F(1, 3162)$ statistic.

7.2.3 Summary of Assumption Evaluation

The following table summarizes all the assumptions required for applying the independent samples t-test and the results of each diagnostic check.

Assumption	Evidence	Result
Normality	SPSS tests significant; Q–Q plots acceptable	Acceptable
Variance Equality	Levene’s Test $p < .001$	Violated → Use Welch’s t-test
Outliers	$< 5\%$ in both groups	Acceptable
Sample Size	$n = 3,164$	Satisfied

Final Decision

Based on the assumption checks and the SPSS evidence:

Welch’s Independent Samples t-Test (One-Tailed) is the appropriate model for comparing permit durations.

This model is robust to unequal variances and moderate non-normality, making it the correct choice for evaluating the research hypothesis.

Chapter 8 Model Evaluation

This chapter presents a rigorous evaluation of the statistical model selected in Chapter 7:

Welch's Independent Samples t-Test. The evaluation includes descriptive statistics, hypothesis testing results, confidence intervals, effect sizes, and an assessment of how well the model addresses the research question. Strengths, limitations, and implications are also discussed to provide a comprehensive understanding of the model's performance.

8.1 Descriptive Statistics

Descriptive statistics summarize the central tendency and variability of permit durations for each group.

Group	N	Mean Duration (days)	Std. Deviation
Residential	2,840	103.27	95.06
Commercial	324	121.66	86.31

Interpretation

Commercial permits take, on average, **18.39 days longer** than residential permits. This difference is the basis for the inferential test conducted below.

8.2 Welch's Independent Samples t-Test

Because Levene's Test indicated unequal variances ($F(1, 3162) = 20.576, p < .001$), the **"Equal variances not assumed"** row in SPSS is the correct model output.

Independent Samples Test									
		t-test for Equality of Means							
		t	df	Significance		Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
				One-Sided p	Two-Sided p			Lower	Upper
duration_days	Equal variances assumed	-3.330	3162	<.001	<.001	-18.394	5.524	-29.224	-7.563
	Equal variances not assumed	-3.595	417.663	<.001	<.001	-18.394	5.116	-28.450	-8.337

Interpretation

- The p-value is **below $\alpha = 0.05$** , indicating a **statistically significant** difference.
- The negative t-value reflects that **Commercial > Residential** in duration.
- The adjusted degrees of freedom (417.663) confirm the use of Welch's correction.

8.3 Confidence Interval Analysis

The 95% confidence interval for the mean difference is:

$$[-28.450, -8.337]$$

Interpretation

We are 95% confident that commercial permits take **between 8 and 29 days longer** than residential permits.

This interval does **not** include zero, reinforcing the statistical significance of the result.

8.4 Effect Size Evaluation

SPSS effect size output provides three standardized measures:

Independent Samples Effect Sizes					
		Standardizer ^a	Point Estimate	95% Confidence Interval	
duration_days	Cohen's d	94.201	-.195	-.310	-.080
	Hedges' correction	94.224	-.195	-.310	-.080
	Glass's delta	86.313	-.213	-.329	-.097

a. The denominator used in estimating the effect sizes.

Cohen's d uses the pooled standard deviation.

Hedges' correction uses the pooled standard deviation, plus a correction factor.

Glass's delta uses the sample standard deviation of the control (i.e., the second) group.

Interpretation

- Cohen's d = **-0.195**, which is a **small effect size**.
- The confidence interval for d does not cross zero, confirming a **real but modest** standardized difference.
- Glass's delta is slightly larger (-0.213), reflecting the smaller SD in the commercial group.

Meaning

Even though the standardized effect is small, the **absolute difference of 18 days** is operationally meaningful for:

- scheduling
- resource allocation
- inspection planning

Effect sizes complement the t-test by showing **how much** the groups differ, not just whether they differ.

8.5 Evaluation Against Benchmarks and Criteria

The model was evaluated against three criteria:

1. Statistical Significance

Achieved

$p < .001$ indicates a reliable difference between groups.

2. Practical Significance

Achieved

An 18-day delay in commercial permits has real operational impact.

3. Model Appropriateness

Achieved

Welch's t-test is the correct model because:

- Variances are unequal
- Sample sizes differ
- Normality is acceptable with large n
- Outliers are minimal

The model meets all methodological requirements.

8.6 Strengths of the Model

1. **Large sample size** ($n = 3,164$) increases reliability
2. **Welch's correction** handles unequal variances
3. **Confidence intervals** provide uncertainty bounds
4. **Effect sizes** quantify magnitude
5. **Real administrative data** increases external validity
6. **Clear directional hypothesis** improves interpretability

8.7 Limitations of the Model

1. Moderate right skew in both groups

2. Administrative data may contain reporting inconsistencies
3. Effect size is small in standardized terms
4. Only two categories (Residential vs. Commercial) were analyzed
5. Does not account for other predictors (e.g., project value, contractor, season)

8.8 Implications for the Research Question

The model provides strong evidence that:

Commercial permits take significantly longer to close than residential permits.

This has several implications:

1. The city may need to allocate more resources to commercial inspections
2. Contractors may require additional oversight
3. Policy adjustments could target bottleneck inspection types
4. Forecasting models can incorporate the 18-day difference for planning

8.9 Conclusion

The model evaluation confirms that Welch's independent samples t-test was the most appropriate analytical approach for this study, given the unequal variances and differing sample sizes across permit categories. The test results provide strong statistical support for the research hypothesis, indicating that permit durations vary significantly between residential and commercial project types. This difference is not only statistically significant but also operationally meaningful, revealing clear patterns in workflow complexity and inspection demand. These findings offer valuable insights for municipal decision-makers, particularly in areas such as staffing, scheduling, and process optimization. By identifying where delays are most likely to occur, the analysis supports evidence-based improvements to Calgary's permitting and inspection processes. While the model is robust, future work could incorporate additional variables, such as contractor performance or seasonal effects, to further refine operational planning. Overall, the evaluation demonstrates that the selected statistical model is both valid and practically impactful for understanding and improving municipal inspection operations.

Appendix A Tables

Table A1. Inspection outcomes by job type,
showing the frequency of each outcome category across all permit types.

Job_Type_Description	Inspection_Outcome	Frequency
(unknown)	Acceptable Add Inspection	6
(unknown)	Not Acceptable	2
Commercial / Multi Family Project	Acceptable Add Inspection	8826
Commercial / Multi Family Project	Not Acceptable	3872
Commercial / Multi Family Project	Acceptable	3718
Commercial / Multi Family Project	Acceptable with Conditions	2436
Commercial / Multi Family Project	Acceptable with Minor Deficiencies	35
Commercial / Multi Family Project	No Inspection	20
Commercial / Multi Family Project	Not Required	16
Commercial / Multi Family Project	Incorrectly called for	13
Commercial / Multi Family Project	Stop work	12
Commercial / Multi Family Project	Unable to enter	11
Commercial / Multi Family Project	CSR Skipped	5
Commercial / Multi Family Project	Continue Monitoring	3
Commercial / Multi Family Project	Cancelled	2
Commercial / Multi Family Project	Blanket Permit Occupancy Acceptable	1
Residential Improvement Project	Acceptable - Close Permit	9046
Residential Improvement Project	Acceptable with Conditions	7584
Residential Improvement Project	Not Acceptable	6948
Residential Improvement Project	Acceptable Add Inspection	2361

Residential Improvement Project	Acceptable	1851
Residential Improvement Project	Acceptable with Minor Deficiencies	88
Residential Improvement Project	Unable to enter	44
Residential Improvement Project	No Inspection	42
Residential Improvement Project	Office Procedure	29
Residential Improvement Project	Stop work	14
Residential Improvement Project	Not Required	11
Residential Improvement Project	Incorrectly called for	8
Residential Improvement Project	Cancelled By Contractor	3
Residential Improvement Project	Cancelled	2
Residential Improvement Project	Covered - Unable to Inspect	1
Residential Improvement Project	CSR Skipped	1
Residential Improvement Project	Charge Re-Inspection	1
Single Construction Permit	Not Acceptable	27970
Single Construction Permit	Acceptable with Conditions	23947
Single Construction Permit	Acceptable	6686
Single Construction Permit	Not Required	1274
Single Construction Permit	Office Procedure	95
Single Construction Permit	Cancelled By Contractor	43
Single Construction Permit	Cancelled	19
Single Construction Permit	No Inspection	7
Single Construction Permit	Acceptable with Minor Deficiencies	4
Single Construction Permit	Unable to enter	4
Single Construction Permit	Acceptable Add Inspection	2
Single Construction Permit	Incorrectly called for	1
Single Construction Permit	Continue Monitoring	1
Single Construction Permit	No Entry	1
Single Construction Permit	Covered - Unable to Inspect	1

Table A2. Outlier Counts and Percentages Across Three-Year Rolling Windows (2000–2025) for Residential and Commercial Permit Durations

Years	Project_Category	Outlier_Count	Total_Records	Outlier_Percentage
2000-2002	Residential	3	20	15
2000-2002	Commercial	1	30	3.33
2001-2003	Residential	3	23	13.04
2001-2003	Commercial	2	30	6.67
2002-2004	Residential	0	22	0
2002-2004	Commercial	4	40	10
2003-2005	Residential	40	998	4.01
2003-2005	Commercial	10	868	1.15
2004-2006	Residential	85	2109	4.03
2004-2006	Commercial	42	1852	2.27
2005-2007	Residential	161	3123	5.16
2005-2007	Commercial	82	2652	3.09
2006-2008	Residential	264	3322	7.95
2006-2008	Commercial	149	2744	5.43
2007-2009	Residential	334	3358	9.95
2007-2009	Commercial	221	2731	8.09
2008-2010	Residential	335	3438	9.74
2008-2010	Commercial	272	2828	9.62
2009-2011	Residential	288	3461	8.32
2009-2011	Commercial	240	3021	7.94
2010-2012	Residential	239	3598	6.64
2010-2012	Commercial	188	3309	5.68
2011-2013	Residential	179	3616	4.95
2011-2013	Commercial	196	3532	5.55
2012-2014	Residential	205	4045	5.07
2012-2014	Commercial	203	3744	5.42
2013-2015	Residential	254	4745	5.35
2013-2015	Commercial	187	4005	4.67
2014-2016	Residential	294	5340	5.51
2014-2016	Commercial	206	4079	5.05
2015-2017	Residential	263	4180	6.29
2015-2017	Commercial	159	2746	5.79
2016-2018	Residential	129	2211	5.83
2016-2018	Commercial	107	1223	8.75
2017-2019	Residential	40	508	7.87
2017-2019	Commercial	3	22	13.64
2018-2020	Residential	186	2399	7.75
2018-2020	Commercial	4	30	13.33
2019-2021	Residential	255	4955	5.15

2019-2021	Commercial	3	47	6.38
2020-2022	Residential	415	7502	5.53
2020-2022	Commercial	11	353	3.12
2021-2023	Residential	208	5407	3.85
2021-2023	Commercial	10	345	2.9
2022-2024	Residential	53	2840	1.87
2022-2024	Commercial	5	324	1.54
2023-2025	Residential	9	286	3.15
2023-2025	Commercial	2	14	14.29

Appendix B SQL Scripts

Script B1. SQL query used to compute row counts across all tables in the Calgary Inspections database.

```
SELECT 'contractor' AS 'Table', COUNT(DISTINCT contractor_id) AS 'Row Counts' FROM  
contractor UNION ALL
```

```
SELECT 'deficiency' AS 'Table', COUNT(DISTINCT deficiency_id) AS 'Row Counts' FROM  
deficiency UNION ALL
```

```
SELECT 'dim_checklist' AS 'Table', COUNT(DISTINCT checklist_id) AS 'Row Counts'  
FROM dim_checklist UNION ALL
```

```
SELECT 'dim_discipline' AS 'Table', COUNT(DISTINCT discipline_id) AS 'Row Counts'  
FROM dim_discipline UNION ALL
```

```
SELECT 'dim_inspection_type' AS 'Table', COUNT(DISTINCT inspection_type_id) AS 'Row  
Counts' FROM dim_inspection_type UNION ALL
```

```
SELECT 'dim_job_type' AS 'Table', COUNT(DISTINCT job_type_id) AS 'Row Counts'  
FROM dim_job_type UNION ALL
```

```
SELECT 'dim_outcome' AS 'Table', COUNT(DISTINCT outcome_id) AS 'Row Counts'  
FROM dim_outcome UNION ALL
```

```
SELECT 'dim_permit_status' AS 'Table', COUNT(DISTINCT permit_status_id) AS 'Row  
Counts' FROM dim_permit_status UNION ALL
```

```
SELECT 'inspection' AS 'Table', COUNT(DISTINCT inspection_id) AS 'Row Counts' FROM  
inspection UNION ALL
```

```
SELECT 'inspection_deficiency' AS 'Table', COUNT(DISTINCT inspection_deficiency_id)  
AS 'Row Counts' FROM inspection_deficiency UNION ALL
```

```
SELECT 'location' AS 'Table', COUNT(DISTINCT location_id) AS 'Row Counts' FROM  
location UNION ALL
```

```
SELECT 'permit' AS 'Table', COUNT(DISTINCT permit_id) AS 'Row Counts' FROM permit  
UNION ALL
```

```
SELECT 'permit_holder' AS 'Table', COUNT(DISTINCT permit_holder_id) AS 'Row Counts'  
FROM permit_holder;
```

Script B2. SQL queries used to extract DISTINCT records from all tables in the Calgary Inspections database.

```
SELECT DISTINCT * FROM contractor;
```

```
SELECT DISTINCT * FROM deficiency;
```

```
SELECT DISTINCT * FROM dim_checklist;
```

```
SELECT DISTINCT * FROM dim_discipline;
SELECT DISTINCT * FROM dim_inspection_type;
SELECT DISTINCT * FROM dim_job_type;
SELECT DISTINCT * FROM dim_outcome;
SELECT DISTINCT * FROM dim_permit_status;
SELECT DISTINCT * FROM inspection;
SELECT DISTINCT * FROM inspection_deficiency;
SELECT DISTINCT * FROM location;
SELECT DISTINCT * FROM permit;
SELECT DISTINCT * FROM permit_holder;
```

Script B3. SQL statements used to document table structures using DESCRIBE across all tables in the Calgary Inspections database.

```
DESCRIBE contractor;
DESCRIBE deficiency;
DESCRIBE dim_checklist;
DESCRIBE dim_discipline;
DESCRIBE dim_inspection_type;
DESCRIBE dim_job_type;
DESCRIBE dim_outcome;
DESCRIBE dim_permit_status;
DESCRIBE inspection;
DESCRIBE inspection_deficiency;
DESCRIBE location;
DESCRIBE permit;
DESCRIBE permit_holder;
```

Script B4. SQL queries used to quantify missing values across all tables in the Calgary Inspections database.

```
SELECT 'contractor' as 'Table',SUM(CASE WHEN contractor_id IS NULL THEN 1 ELSE 0
END) AS contractor_id, SUM(CASE WHEN contractor_name IS NULL THEN 1 ELSE 0
END) AS contractor_name FROM contractor;
```

```
SELECT 'deficiency' as 'Table', SUM(CASE WHEN deficiency_id IS NULL THEN 1 ELSE
0 END) AS deficiency_id, SUM(CASE WHEN deficiency_text IS NULL THEN 1 ELSE 0
END) AS deficiency_text FROM deficiency;
```

```
SELECT 'dim_checklist' as 'Table', SUM(CASE WHEN checklist_id IS NULL THEN 1
ELSE 0 END) AS checklist_id, SUM(CASE WHEN checklist_name IS NULL THEN 1
ELSE 0 END) AS checklist_name FROM dim_checklist;
```

```
SELECT 'dim_discipline' as 'Table', SUM(CASE WHEN discipline_id IS NULL THEN 1
ELSE 0 END) AS discipline_id, SUM(CASE WHEN discipline_desc IS NULL THEN 1
ELSE 0 END) AS discipline_desc FROM dim_discipline;
```

```
SELECT 'dim_inspection_type' as 'Table', SUM(CASE WHEN inspection_type_id IS NULL
THEN 1 ELSE 0 END) AS inspection_type_id,SUM(CASE WHEN inspection_type_desc IS
NULL THEN 1 ELSE 0 END) AS inspection_type_desc FROM dim_inspection_type;
```

```
SELECT 'dim_job_type' as 'Table', SUM(CASE WHEN job_type_id IS NULL THEN 1
ELSE 0 END) AS job_type_id, SUM(CASE WHEN job_type_desc IS NULL THEN 1 ELSE
0 END) AS job_type_desc FROM dim_job_type;
```

```
SELECT 'dim_outcome' as 'Table', SUM(CASE WHEN outcome_id IS NULL THEN 1 ELSE
0 END) AS outcome_id, SUM(CASE WHEN outcome_desc IS NULL THEN 1 ELSE 0
END) AS outcome_desc FROM dim_outcome;
```

```
SELECT 'dim_permit_status' as 'Table', SUM(CASE WHEN permit_status_id IS NULL
THEN 1 ELSE 0 END) AS permit_status_id, SUM(CASE WHEN status_name IS NULL
THEN 1 ELSE 0 END) AS status_name FROM dim_permit_status;
```

```
SELECT 'inspection' as 'Table',SUM(CASE WHEN inspection_id IS NULL THEN 1 ELSE 0
END) AS inspection_id,SUM(CASE WHEN permit_id IS NULL THEN 1 ELSE 0 END) AS
permit_id,SUM(CASE WHEN process_id IS NULL THEN 1 ELSE 0 END) AS
process_id,SUM(CASE WHEN inspection_type_id IS NULL THEN 1 ELSE 0 END) AS
inspection_type_id,SUM(CASE WHEN outcome_id IS NULL THEN 1 ELSE 0 END) AS
outcome_id,SUM(CASE WHEN inspection_completed_date IS NULL THEN 1 ELSE 0
END) AS inspection_completed_date,SUM(CASE WHEN permit_deficiencies_flag IS NULL
THEN 1 ELSE 0 END) AS permit_deficiencies_flag FROM inspection;
```

```
SELECT 'inspection_deficiency' as 'Table',SUM(CASE WHEN inspection_deficiency_id IS
NULL THEN 1 ELSE 0 END) AS inspection_deficiency_id,SUM(CASE WHEN
inspection_id IS NULL THEN 1 ELSE 0 END) AS inspection_id,SUM(CASE WHEN
deficiency_id IS NULL THEN 1 ELSE 0 END) AS deficiency_id,SUM(CASE WHEN
discipline_id IS NULL THEN 1 ELSE 0 END) AS discipline_id,SUM(CASE WHEN
checklist_id IS NULL THEN 1 ELSE 0 END) AS checklist_id,SUM(CASE WHEN
deficiency_resolved IS NULL THEN 1 ELSE 0 END) AS deficiency_resolved,SUM(CASE
WHEN resolved_date IS NULL THEN 1 ELSE 0 END) AS resolved_date FROM
inspection_deficiency;
```

```

SELECT 'location' as 'Table',SUM(CASE WHEN location_id IS NULL THEN 1 ELSE 0
END) AS location_id,SUM(CASE WHEN address_line IS NULL THEN 1 ELSE 0 END) AS
address_line,SUM(CASE WHEN longitude IS NULL THEN 1 ELSE 0 END) AS
longitude,SUM(CASE WHEN latitude IS NULL THEN 1 ELSE 0 END) AS
latitude,SUM(CASE WHEN point_wkt IS NULL THEN 1 ELSE 0 END) AS point_wkt
FROM location;

```

```

SELECT 'permit' as 'Table',SUM(CASE WHEN permit_id IS NULL THEN 1 ELSE 0 END)
AS permit_id,SUM(CASE WHEN external_file_num IS NULL THEN 1 ELSE 0 END) AS
external_file_num,SUM(CASE WHEN job_type_id IS NULL THEN 1 ELSE 0 END) AS
job_type_id,SUM(CASE WHEN location_id IS NULL THEN 1 ELSE 0 END) AS
location_id,SUM(CASE WHEN permit_status_id IS NULL THEN 1 ELSE 0 END) AS
permit_status_id,SUM(CASE WHEN contractor_id IS NULL THEN 1 ELSE 0 END) AS
contractor_id,SUM(CASE WHEN permit_holder_id IS NULL THEN 1 ELSE 0 END) AS
permit_holder_id,SUM(CASE WHEN issue_date IS NULL THEN 1 ELSE 0 END) AS
issue_date,SUM(CASE WHEN permit_close_date IS NULL THEN 1 ELSE 0 END) AS
permit_close_date FROM permit;

```

```

SELECT 'permit_holder' as 'Table',SUM(CASE WHEN permit_holder_id IS NULL THEN 1
ELSE 0 END) AS permit_holder_id,SUM(CASE WHEN holder_name IS NULL THEN 1
ELSE 0 END) AS holder_name FROM permit_holder;

```

Script B5. SQL script used to construct the analytical dataset through data extraction, transformation, and filtering procedures

```

SELECT
    A.permit_id,
    A.job_type_id,
    CASE
        WHEN B.job_type_desc = 'Residential Improvement Project' THEN 'Residential'
        WHEN B.job_type_desc = 'Commercial / Multi Family Project' THEN 'Commercial'
    END AS job_type_description,
    A.permit_status_id,
    C.status_name,
    A.issue_date,
    A.permit_close_date,
    DATEDIFF(A.permit_close_date, A.issue_date) AS permit_duration_days
FROM permit AS A
LEFT JOIN dim_job_type AS B

```

```

    ON A.job_type_id = B.job_type_id
LEFT JOIN dim_permit_status AS C
    ON A.permit_status_id = C.permit_status_id
WHERE A.issue_date IS NOT NULL
    AND A.permit_close_date IS NOT NULL
    AND A.permit_status_id = 1
    AND YEAR(issue_date) BETWEEN 2022 AND 2024
    AND A.job_type_id IN (2, 3);

```

Script B6. SQL query used to generate three-year rolling windows and identify outliers across Residential and Commercial sectors

```

WITH RECURSIVE YearWindows AS (
    -- Anchor: first 3-year window
    SELECT 2000 AS start_year, 2002 AS end_year

    UNION ALL

    -- Recursive: shift window forward by 1 year until 2025
    SELECT start_year + 1, end_year + 1
    FROM YearWindows
    WHERE end_year < 2025
),
FilteredPermits AS (
    SELECT
        CASE
            WHEN job_type_id = 3 THEN 'Commercial'
            WHEN job_type_id = 2 THEN 'Residential'
            ELSE 'Other'
        END AS project_category,
        YEAR(issue_date) AS issue_year,
        DATEDIFF(permit_close_date, issue_date) AS duration_days
    FROM permit

```

```

WHERE issue_date >= '2000-01-01'

AND issue_date <= '2025-12-31'

AND permit_close_date IS NOT NULL

AND YEAR(permit_close_date) < 2262
),
ValidGroups AS (
SELECT *
FROM FilteredPermits
WHERE project_category != 'Other'
),
WindowedPermits AS (
-- Assign permits to each rolling window
SELECT
    w.start_year,
    w.end_year,
    CONCAT(w.start_year, '-', w.end_year) AS time_window,
    p.project_category,
    p.duration_days,
    ROW_NUMBER() OVER (
        PARTITION BY w.start_year, p.project_category
        ORDER BY p.duration_days ASC
    ) AS row_num,
    COUNT(*) OVER (
        PARTITION BY w.start_year, p.project_category
    ) AS total_rows
FROM YearWindows w
JOIN ValidGroups p
    ON p.issue_year BETWEEN w.start_year AND w.end_year
),

```

```

WindowedQuartiles AS (
  -- Compute Q1 and Q3 using linear interpolation
  SELECT
    time_window,
    start_year,
    end_year,
    project_category,
    -- Q1
    MAX(CASE WHEN row_num = FLOOR(1 + 0.25 * (total_rows - 1))
      THEN duration_days END)
      * (1 - ((1 + 0.25 * (total_rows - 1))
        - FLOOR(1 + 0.25 * (total_rows - 1))))
    +
    MAX(CASE WHEN row_num = CEIL(1 + 0.25 * (total_rows - 1))
      THEN duration_days END)
      * ((1 + 0.25 * (total_rows - 1))
        - FLOOR(1 + 0.25 * (total_rows - 1))) AS q1,
    -- Q3
    MAX(CASE WHEN row_num = FLOOR(1 + 0.75 * (total_rows - 1))
      THEN duration_days END)
      * (1 - ((1 + 0.75 * (total_rows - 1))
        - FLOOR(1 + 0.75 * (total_rows - 1))))
    +
    MAX(CASE WHEN row_num = CEIL(1 + 0.75 * (total_rows - 1))
      THEN duration_days END)
      * ((1 + 0.75 * (total_rows - 1))
        - FLOOR(1 + 0.75 * (total_rows - 1))) AS q3
  FROM WindowedPermits
  GROUP BY time_window, start_year, end_year, project_category
),

```



```

WindowedThresholds AS (
  -- Compute outlier fences
  SELECT
    time_window,
    start_year,
    end_year,
    project_category,
    (q1 - 1.5 * (q3 - q1)) AS lower_bound,
    (q3 + 1.5 * (q3 - q1)) AS upper_bound
  FROM WindowedQuartiles
)
-- Final Output
SELECT
  t.time_window AS Years,
  t.project_category AS Project_Category,
  COUNT(CASE WHEN wp.duration_days < t.lower_bound
    OR wp.duration_days > t.upper_bound
  THEN 1 END) AS Outlier_Count,
  COUNT(*) AS Total_Records,
  ROUND(
    COUNT(CASE WHEN wp.duration_days < t.lower_bound
    OR wp.duration_days > t.upper_bound
  THEN 1 END) * 100.0 / COUNT(),
    2
  ) AS Outlier_Percentage
FROM WindowedPermits wp
JOIN WindowedThresholds t
  ON wp.time_window = t.time_window
  AND wp.project_category = t.project_category
GROUP BY

```

```
t.start_year,  
t.end_year,  
t.time_window,  
t.project_category,  
t.lower_bound,  
t.upper_bound
```

ORDER BY

```
t.start_year ASC,  
t.project_category DESC;
```

Script B7. SQL for Mean, Median, Standard Deviation, and Range (2022–2024 Window)

WITH FilteredPermits **AS** (

SELECT

CASE

WHEN job_type_id = 3 **THEN** 'Commercial'

WHEN job_type_id = 2 **THEN** 'Residential'

ELSE 'Other'

END AS project_category,

DATEDIFF(permit_close_date, issue_date) **AS** duration_days

FROM permit

WHERE issue_date >= '2022-01-01'

AND issue_date <= '2024-12-31'

AND permit_close_date **IS NOT NULL**

),

ValidGroups **AS** (

-- Excludes job_type_id 1 and 4 automatically

SELECT *

FROM FilteredPermits

WHERE project_category != 'Other'

),

```

NumberedPermits AS (
    SELECT
        project_category,
        duration_days,
        ROW_NUMBER() OVER (
            PARTITION BY project_category
            ORDER BY duration_days ASC
        ) AS row_num,
        COUNT(*) OVER (
            PARTITION BY project_category
        ) AS total_rows
    FROM ValidGroups
)
SELECT
    project_category,
    COUNT(*) AS `count`,
    ROUND(AVG(duration_days), 6) AS `mean`,

-- Median using linear interpolation
ROUND(
    MAX(
        CASE WHEN row_num = FLOOR(1 + 0.5 * (total_rows - 1))
        THEN duration_days END
    ) * (1 - ((1 + 0.5 * (total_rows - 1))
        - FLOOR(1 + 0.5 * (total_rows - 1))))
    +
    MAX(
        CASE WHEN row_num = CEIL(1 + 0.5 * (total_rows - 1))
        THEN duration_days END
    ) * ((1 + 0.5 * (total_rows - 1))

```

```

        - FLOOR(1 + 0.5 * (total_rows - 1))),
    1
) AS `median`,
ROUND(STDDEV_SAMP(duration_days), 6) AS `std`,
MIN(duration_days) AS `min`,
MAX(duration_days) AS `max`

FROM NumberedPermits
GROUP BY project_category;

Script B8. SQL for Quartiles, Outlier Counts, and Outlier Percentages (2022–2024 Window)
WITH FilteredPermits AS (
    SELECT
        CASE
            WHEN job_type_id = 3 THEN 'Commercial'
            WHEN job_type_id = 2 THEN 'Residential'
            ELSE 'Other'
        END AS project_category,
        DATEDIFF(permit_close_date, issue_date) AS duration_days
    FROM permit
    WHERE issue_date >= '2022-01-01'
        AND issue_date <= '2024-12-31'
        AND permit_close_date IS NOT NULL
        AND YEAR(permit_close_date) < 2262
),

ValidGroups AS (
    -- Excludes job_type_id 1 and 4 automatically
    SELECT *
    FROM FilteredPermits

```

```

WHERE project_category != 'Other'
),

```

```

NumberedPermits AS (
  SELECT
    project_category,
    duration_days,
    ROW_NUMBER() OVER (
      PARTITION BY project_category
      ORDER BY duration_days ASC
    ) AS row_num,
    COUNT(*) OVER (
      PARTITION BY project_category
    ) AS total_rows
  FROM ValidGroups
),

```

```

Quartiles AS (
  SELECT
    project_category,

    -- Q1 using linear interpolation
    MAX(
      CASE WHEN row_num = FLOOR(1 + 0.25 * (total_rows - 1))
      THEN duration_days END
    ) * (1 - ((1 + 0.25 * (total_rows - 1))
      - FLOOR(1 + 0.25 * (total_rows - 1))))
    +
    MAX(
      CASE WHEN row_num = CEIL(1 + 0.25 * (total_rows - 1))

```

```

        THEN duration_days END
    ) * ((1 + 0.25 * (total_rows - 1))
        - FLOOR(1 + 0.25 * (total_rows - 1))) AS q1,

    -- Q3 using linear interpolation
    MAX(
        CASE WHEN row_num = FLOOR(1 + 0.75 * (total_rows - 1))
            THEN duration_days END
    ) * (1 - ((1 + 0.75 * (total_rows - 1))
        - FLOOR(1 + 0.75 * (total_rows - 1))))
    +
    MAX(
        CASE WHEN row_num = CEIL(1 + 0.75 * (total_rows - 1))
            THEN duration_days END
    ) * ((1 + 0.75 * (total_rows - 1))
        - FLOOR(1 + 0.75 * (total_rows - 1))) AS q3

FROM NumberedPermits
GROUP BY project_category
),

```

```

OutlierThresholds AS (
    SELECT
        project_category,
        q1,
        q3,
        (q3 - q1) AS iqr,
        (q1 - 1.5 * (q3 - q1)) AS lower_bound,
        (q3 + 1.5 * (q3 - q1)) AS upper_bound
    FROM Quartiles

```

)

SELECT

fp.project_category,
ROUND(t.q1, 2) **AS** first_quartile,
ROUND(t.q3, 2) **AS** third_quartile,
ROUND(t.lower_bound, 2) **AS** lower_bound_days,
ROUND(t.upper_bound, 2) **AS** upper_bound_days,

COUNT(
 CASE WHEN fp.duration_days < t.lower_bound
 OR fp.duration_days > t.upper_bound
 THEN 1 END
) **AS** outlier_count,

COUNT(*) **AS** total_records,

ROUND(
 COUNT(
 CASE WHEN fp.duration_days < t.lower_bound
 OR fp.duration_days > t.upper_bound
 THEN 1 END
) * 100.0 / **COUNT**(),
 2
) **AS** outlier_percentage

FROM ValidGroups fp

JOIN OutlierThresholds t

ON fp.project_category = t.project_category

GROUP BY

```
fp.project_category,  
t.q1,  
t.q3,  
t.lower_bound,  
t.upper_bound;
```